

MA2104 Multivariable Calculus

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AY25/26 Sem 2

This is a course on the calculus of functions of several real variables, applications of which abound in mathematics, the physical sciences and engineering. The aim is for students to acquire computational skills, ability for 2- and 3-D visualisation and to understand conceptually fundamental results such as Green's Theorem, Stokes' Theorem and the Divergence Theorem. Major topics: Euclidean distance and elementary topological concepts in $\mathbb{R}^n$, limit and continuity, implicit functions. Partial differentiation, differentiable functions, differentials, chain rules, directional derivatives, gradients, mean value theorem, Taylor's formula, extreme value theorem, Lagrange multipliers. Multiple integrals and iterated integrals, change of order of integration, applications, Jacobian matrix, change of variables in multiple integrals. Line integrals and Green's theorem. Surface integrals, Stokes' Theorem, Divergence Theorem.

Prerequisite

If undertaking an Undergraduate Degree THEN(must have completed 1 of MA1505 / MA1511 / MA1521 / MA2002 / PC2174A at a grade of at least D)

Preclusion

If undertaking an Undergraduate Degree THEN ((must not have completed 1 of MA2311 / YSC2252 at a grade of at least D))

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1 Vectors in $\mathbb{R}^3$

1.1 Vectors

A 3D vector is of the form $\mathbf{u} = \langle a, b, c \rangle$, where a, b, c are called the **components** of the vector.

The vector $\langle a, b, c \rangle$ is called the **position vector** of the point $P(a, b, c)$.

Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$. The two vectors are equal if they are equal component-wise (same direction and same magnitude):

$$\mathbf{a} = \mathbf{b} \iff a_i = b_i \quad \text{for } i = 1, 2, 3.$$

Addition is defined as

$$\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle.$$

Let $\lambda \in \mathbb{R}$. Scalar multiplication is defined as

$$\lambda \mathbf{a} = \langle \lambda a_1, \lambda a_2, \lambda a_3 \rangle.$$

Let $\mathbb{V}^3 = \{ \langle x, y, z \rangle : x, y, z \in \mathbb{R} \}$ denote the set of three-dimensional vectors in $\mathbb{R}^3$. (There exists a bijection from $\mathbb{R}^3$ to $\mathbb{V}^3$.)

Theorem 1.1. $\mathbb{V}^3$ is a vector space over $\mathbb{R}$, with addition and scalar multiplication defined above. That is, let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$, $\alpha, \beta \in \mathbb{R}$. Then

(i) $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$.

(ii) $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$.

(iii) $\mathbf{a} + \mathbf{0} = \mathbf{a}$.

(iv) $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$.

(v) $\alpha(\mathbf{a} + \mathbf{b}) = \alpha\mathbf{a} + \alpha\mathbf{b}$.

(vi) $\alpha\mathbf{a} = \alpha\mathbf{a}$.

(vii) $(\alpha + \beta)\mathbf{a} = \alpha\mathbf{a} + \beta\mathbf{a}$.

(viii) $(\alpha\beta)\mathbf{a} = \alpha(\beta\mathbf{a})$.

(ix) $1\mathbf{a} = \mathbf{a}$.

We represent a vector as a directed line segment $\mathbf{u} = \overrightarrow{AB}$ when we want to specify the initial point $A(a_1, a_2, a_3)$ and terminal point $B(b_1, b_2, b_3)$:

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \langle b_1, b_2, b_3 \rangle - \langle a_1, a_2, a_3 \rangle \\ &= \langle b_1 - a_1, b_2 - a_2, b_3 - a_3 \rangle. \end{aligned}$$

Definition 1.2. The **norm** (or **length**, or **magnitude**) of the vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ is

$$\|\mathbf{a}\| := \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

The canonical basis of $\mathbb{V}^3$ is $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, where $\mathbf{i} = \langle 1, 0, 0 \rangle$, $\mathbf{j} = \langle 0, 1, 0 \rangle$, $\mathbf{k} = \langle 0, 0, 1 \rangle$; this basis is known as the **standard basis**. Hence, any 3D vector can be written as a (unique) linear combination of the standard basis vectors:

$$\langle a, b, c \rangle = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}.$$

Theorem 1.3. Let $\mathbf{a}$ be a vector, and $\alpha \in \mathbb{R}$. Then $\|\alpha\mathbf{u}\| = |\alpha|\|\mathbf{u}\|$.

Proof. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$. Then

$$\begin{aligned}\|\alpha \mathbf{u}\| &= \sqrt{(\alpha u_1)^2 + (\alpha u_2)^2 + (\alpha u_3)^2} \\ &= \sqrt{\alpha^2(u_1^2 + u_2^2 + u_3^2)} \\ &= \sqrt{\alpha^2} \sqrt{u_1^2 + u_2^2 + u_3^2} \\ &= |\alpha| \|\mathbf{u}\|.\end{aligned}$$

□

Definition 1.4. A **unit vector** is a vector with norm 1.

Sometimes to specify that a vector $\mathbf{a}$ is a unit vector, we write $\hat{\mathbf{a}}$.

Theorem 1.5. If $\mathbf{a} \neq \mathbf{0}$, then a unit vector in the same direction as $\mathbf{a}$ is given by

$$\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|}.$$

Proof. Notice $1/\|\mathbf{a}\|$ is a positive scalar, so $\mathbf{u}$ is in the same direction as $\mathbf{a}$. It remains to show that $\mathbf{u}$ is a unit vector, i.e., $\|\mathbf{u}\| = 1$. By the preceding theorem,

$$\|\mathbf{u}\| = \left\| \frac{\mathbf{a}}{\|\mathbf{a}\|} \right\| = \frac{1}{\|\mathbf{a}\|} \|\mathbf{a}\| = 1.$$

Hence, $\mathbf{u}$ is the unit vector in the same direction as $\mathbf{a}$.

□

Theorem 1.6 (Triangle inequality). For any vectors $\mathbf{u}$ and $\mathbf{v}$,

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|. \quad (1)$$

Proof. By the Cauchy–Schwarz inequality, we have

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\mathbf{u} \cdot \mathbf{v} \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2|\mathbf{u} \cdot \mathbf{v}| \\ &\leq \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\|\mathbf{u}\| \|\mathbf{v}\| \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.\end{aligned}$$

Taking square roots on both sides yields the desired inequality.

□

1.2 Dot product

Definition 1.7. Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$. The **dot product** (or **inner product**) of $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{a} \cdot \mathbf{b} := a_1 b_1 + a_2 b_2 + a_3 b_3.$$

Clearly we have

$$\mathbf{i} \cdot \mathbf{j} = \mathbf{i} \cdot \mathbf{k} = \mathbf{j} \cdot \mathbf{k} = 0 \quad \text{and} \quad \mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1.$$

Theorem 1.8 (Properties of dot product). Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$, $\alpha \in \mathbb{R}$. Then

$$(i) \quad \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}. \quad (\text{commutativity})$$

$$(ii) \quad \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}. \quad (\text{distributive law})$$

$$(iii) \quad (\alpha \mathbf{a}) \cdot \mathbf{b} = \alpha(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (\alpha \mathbf{b}).$$

(iv) $\mathbf{0} \cdot \mathbf{a} = 0$.

(v) $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$.

Proof.

(i) This follows from the commutativity of multiplication on $\mathbb{R}$ as a field.

(ii) Distributivity on $\mathbb{R}$

(iii)

(iv)

(v)

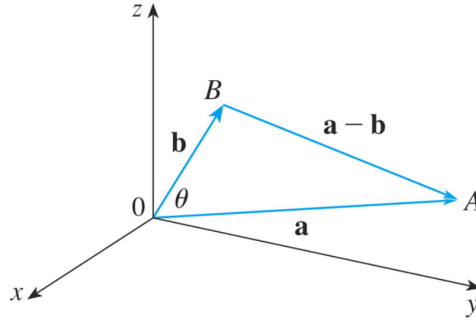
□

For two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$, by convention, we consider the **angle** θ between them to be the smaller angle between $\mathbf{a}$ and $\mathbf{b}$, formed by placing their initial points at the origin.

Theorem 1.9. *Let θ be the angle between non-zero vectors $\mathbf{a}$ and $\mathbf{b}$. Then*

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta. \quad (2)$$

Proof. Let $\mathbf{a} = \overrightarrow{OA}$ and $\mathbf{b} = \overrightarrow{OB}$, where O is the origin and $\theta = \angle AOB$.



Applying cosine rule to $\triangle OAB$,

$$\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta.$$

Note that $\|\mathbf{a} - \mathbf{b}\|^2 = (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = \|\mathbf{a}\|^2 - 2\mathbf{a} \cdot \mathbf{b} + \|\mathbf{b}\|^2$. Rearranging yields the desired result. □

Then the angle between $\mathbf{a}$ and $\mathbf{b}$ is

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}.$$

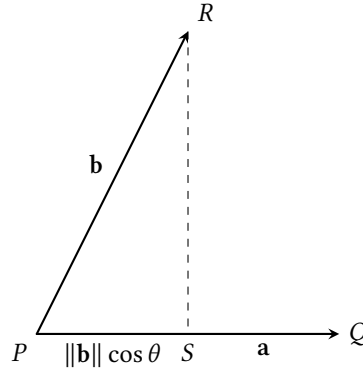
Corollary 1.10. *Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if and only if $\mathbf{a} \cdot \mathbf{b} = 0$.*

Proof.

$$\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta = \mathbf{a} \cdot \mathbf{b} = 0$$

if and only if $\cos \theta = 0$, if and only if $\theta = \frac{\pi}{2}$, which is equivalent to saying that $\mathbf{a}$ and $\mathbf{b}$ are orthogonal. □

Next, we discuss the projection of a vector along another vector. Let $\mathbf{a}$ and $\mathbf{b}$ be two vectors in $\mathbb{R}^3$.



Definition 1.11. The vector $\overrightarrow{PS}$ is the **vector projection** of $\mathbf{b}$ onto $\mathbf{a}$, denoted by $\text{proj}_{\mathbf{a}}\mathbf{b}$.

The **scalar projection** of $\mathbf{b}$ onto $\mathbf{a}$ is the signed magnitude of the vector projection, denoted by $\text{comp}_{\mathbf{a}}\mathbf{b}$.

Notice that $\text{comp}_{\mathbf{a}}\mathbf{b}$ is the (directed) length of the line segment PS :

$$\text{comp}_{\mathbf{a}}\mathbf{b} = \|\mathbf{b}\| \cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|}$$

by applying Theorem 1.9. Hence, the vector projection of $\mathbf{b}$ onto $\mathbf{a}$ is the scalar projection times the unit vector in the direction of $\mathbf{a}$:

$$\text{proj}_{\mathbf{a}}\mathbf{b} = (\text{comp}_{\mathbf{a}}\mathbf{b}) \frac{\mathbf{a}}{\|\mathbf{a}\|} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} \right) \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a}.$$

Theorem 1.12 (Cauchy–Schwarz inequality). *For any vectors $\mathbf{u}$ and $\mathbf{v}$,*

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|. \quad (3)$$

Theorem 1.13 (Parallelogram law).

$$\|\mathbf{a} + \mathbf{b}\|^2 + \|\mathbf{a} - \mathbf{b}\|^2 = 2\|\mathbf{a}\|^2 + 2\|\mathbf{b}\|^2. \quad (4)$$

Proof. Evaluate

$$\begin{aligned} \|\mathbf{a} + \mathbf{b}\|^2 + \|\mathbf{a} - \mathbf{b}\|^2 &= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) + (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) \\ &= \|\mathbf{a}\|^2 + 2\mathbf{a} \cdot \mathbf{b} + \|\mathbf{b}\|^2 + \|\mathbf{a}\|^2 - 2\mathbf{a} \cdot \mathbf{b} + \|\mathbf{b}\|^2 \\ &= 2\|\mathbf{a}\|^2 + 2\|\mathbf{b}\|^2. \end{aligned}$$

□

1.3 Cross product

Definition 1.14. Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$. The **cross product** of $\mathbf{a}$ and $\mathbf{b}$ is

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \end{aligned}$$

For brevity of notation, sometimes the above determinant is written as

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a & b & c \\ b & c & a \end{vmatrix}.$$

Clearly we have

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \quad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}, \quad \mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0}.$$

Theorem 1.15 (Properties of cross product). *Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be vectors in $\mathbb{R}^3$. Then*

$$(i) \quad \mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$

$$(ii) \quad (\alpha \mathbf{a}) \times \mathbf{b} = \alpha(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (\alpha \mathbf{b})$$

$$(iii) \quad \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$$

$$(iv) \quad (\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$$

$$(v) \quad \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$$

$$(vi) \quad \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$$

Proof. These follow from the definition and properties of the determinant. For (i),

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = - \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = -\mathbf{b} \times \mathbf{a}.$$

□

Theorem 1.16. *Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$. Then*

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}. \quad (5)$$

We call $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ the **scalar triple product** of $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$.

Proof. Compute

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= a_1(b_2c_3 - b_3c_2) + a_2(b_3c_1 - b_1c_3) + a_3(b_1c_2 - b_2c_1) \\ &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}. \end{aligned}$$

□

Corollary 1.17. $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

Theorem 1.18. *If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, where $0 \leq \theta \leq \pi$, then*

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta. \quad (6)$$

Proof. First we need the following identity:

$$(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 = (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2$$

which can be easily verified by direct simplification of both sides.

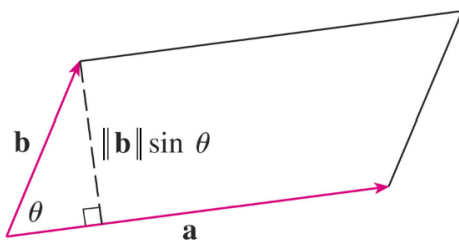
Using this identity, we have

$$\begin{aligned}
\|\mathbf{a} \times \mathbf{b}\|^2 &= (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \\
&= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \\
&= \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2 \\
&= \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \|\mathbf{a}\|^2\|\mathbf{b}\|^2 \cos^2 \theta \\
&= \|\mathbf{a}\|^2\|\mathbf{b}\|^2 \sin^2 \theta.
\end{aligned}$$

Since $\sin \theta \geq 0$ when $0 \leq \theta \leq \pi$, taking square roots on both sides yields the desired result. $\square$

Corollary 1.19. $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

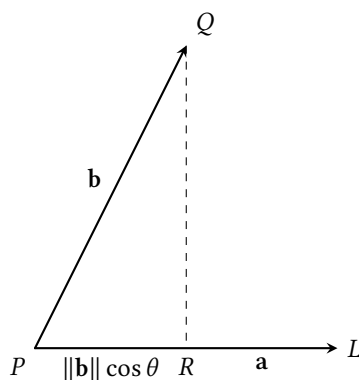
Consider the parallelogram spanned by the vectors $\mathbf{a}$ and $\mathbf{b}$:



The area of this parallelogram is

$$\begin{aligned}
\text{Area} &= \text{Height} \times \text{Base} \\
&= (\|\mathbf{b}\| \sin \theta) \|\mathbf{a}\| \\
&= \|\mathbf{a} \times \mathbf{b}\|.
\end{aligned}$$

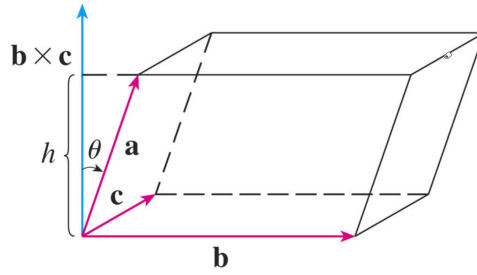
Consider finding the distance from a point Q to a line L . Let P and R be two points on L .



Let θ be the angle between $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. Then the distance is

$$\|\overrightarrow{PQ}\| \sin \theta = \frac{\|\overrightarrow{PQ} \times \overrightarrow{PR}\|}{\|\overrightarrow{PR}\|}.$$

Consider the parallelepiped spanned by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$:



Let θ be the angle between $\mathbf{a}$ and $\mathbf{b} \times \mathbf{c}$. Then the volume of this parallelepiped is

$$\begin{aligned} \text{Volume} &= \text{Base area} \times \text{Height} \\ &= \|\mathbf{b} \times \mathbf{c}\| \|\mathbf{a}\| \cos \theta \\ &= |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|. \end{aligned}$$

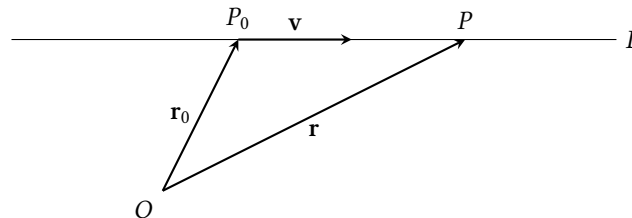
Corollary 1.20. $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar if and only if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$.

1.4 Lines

To locate a particular line L in space, we need

1. a point $P_0(x_0, y_0, z_0)$ on the line L
2. a vector $\mathbf{v}$ whose direction is parallel to the line L .

Let L be a line passing through a point $P_0(x_0, y_0, z_0)$ in the direction of the vector $\mathbf{v} = \langle a, b, c \rangle$. Then any point P on L has position vector $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ for some $t \in \mathbb{R}$.



The **vector equation** of a line is

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}.$$

Write $\mathbf{r} = \langle x, y, z \rangle$. Then $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ is equivalent to $\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle$. By comparing components, this gives the **parametric equation** of a line:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$$

Eliminating t , we obtain **symmetric equations** of a line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Let L_1 and L_2 be two lines in $\mathbb{R}^3$, with direction vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. Let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$.

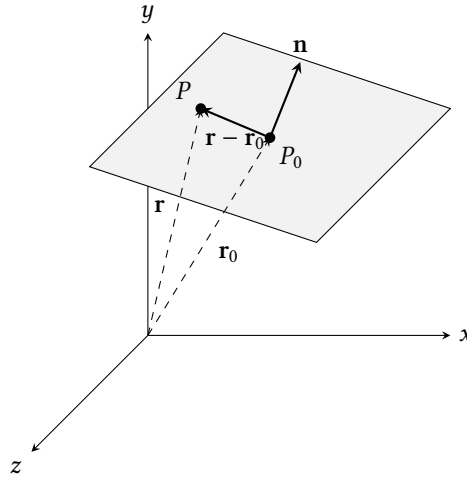
- The lines L_1 and L_2 are **parallel** whenever $\mathbf{a}$ and $\mathbf{b}$ are parallel.
- If L_1 and L_2 **intersect**, then θ is an angle between L_1 and L_2 . Notice $\pi - \theta$ is also an angle between the lines.
- The lines are **skew** if they are non-parallel and non-intersecting.

1.5 Planes

To locate a particular plane Π in space, we need

1. a point $P_0(x_0, y_0, z_0)$ on the plane Π
2. a vector $\mathbf{n}$ whose direction is perpendicular to the plane Π .

Consider a plane Π in $\mathbb{R}^3$ passing through a point $P_0(x_0, y_0, z_0)$ with normal vector $\mathbf{n}$. Let $P(x, y, z)$ be an arbitrary point on Π . Let $\mathbf{r}$ and $\mathbf{r}_0$ be the position vectors of P and P_0 respectively.



Since $\overrightarrow{P_0P} = \mathbf{r} - \mathbf{r}_0$ lies on the plane, it is perpendicular to $\mathbf{n}$. Hence,

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0,$$

called the **vector equation** of the plane.

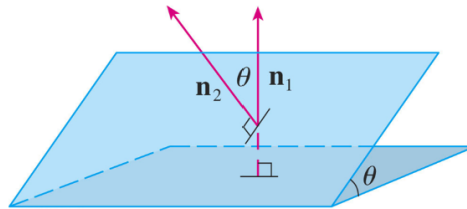
In terms of components, $\mathbf{n} = \langle a, b, c \rangle$, $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$. Then the vector equation can be written as

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

In general, a linear equation in x, y, z , i.e., $ax + by + cz + d = 0$ is an equation of a plane in $\mathbb{R}^3$; note that $\langle a, b, c \rangle$ is a normal vector to the plane.

Remark. Given vectors on the plane, a normal vector $\mathbf{n}$ can be found by taking the cross product.

Two planes are **parallel** if their normal vectors are parallel. If two planes are not parallel, then they intersect in a straight line. By angle chasing, the angle between the two planes is the angle θ between their normal vectors:



Proposition 1.21. The distance between two parallel planes $\Pi_1 : ax + by + cz + d_1 = 0$ and $\Pi_2 : ax + by + cz + d_2 = 0$ is

$$D = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.$$

Proof. For $j = 1, 2$, take a point $P_j = (x_j, y_j, z_j)$ from Π_j , so

$$ax_j + by_j + cz_j + d_j = 0.$$

The distance D is given by the scalar projection of the vector $\overrightarrow{P_1P_2} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$ onto the common normal vector $\mathbf{n} = \langle a, b, c \rangle$:

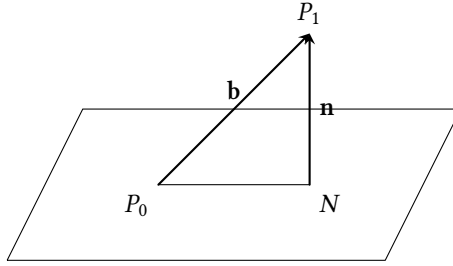
$$\begin{aligned}
 D &= \text{comp}_{\mathbf{n}} \overrightarrow{P_1P_2} \\
 &= \frac{|\overrightarrow{P_1P_2} \cdot \mathbf{n}|}{\|\mathbf{n}\|} \\
 &= \frac{|(ax_2 + by_2 + cz_2) - (ax_1 + by_1 + cz_1)|}{\sqrt{a^2 + b^2 + c^2}} \\
 &= \frac{|-d_2 - (-d_1)|}{\sqrt{a^2 + b^2 + c^2}} \\
 &= \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.
 \end{aligned}$$

□

Proposition 1.22. The distance from a point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is

$$\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Proof. Pick a point $P_0(x_0, y_0, z_0)$ on the plane. Let $\mathbf{b} = \overrightarrow{P_0P_1} = \langle x_1 - x_0, y_1 - y_0, z_1 - z_0 \rangle$.



Then

$$\begin{aligned}
 \|\overrightarrow{NP_1}\| &= \|\text{proj}_{\mathbf{n}} \mathbf{b}\| = \frac{|\mathbf{n} \cdot \mathbf{b}|}{\|\mathbf{n}\|} \\
 &= \frac{|a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0)|}{\sqrt{a^2 + b^2 + c^2}} \\
 &= \frac{|(ax_1 + by_1 + cz_1) - (ax_0 + by_0 + cz_0)|}{\sqrt{a^2 + b^2 + c^2}} \\
 &= \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}
 \end{aligned}$$

where the last line follows since P_0 lies on the plane, so its coordinates satisfy the equation of the plane, i.e., $ax_0 + by_0 + cz_0 + d = 0$. □

To find the distance between two parallel planes (normal vectors are parallel), pick any point on one plane, then apply above result.

2 Cylindrical and Spherical Coordinates

2.1 Polar coordinates

Given a point P with Cartesian coordinates (x, y) , we can locate its position using its distance r from the origin, and the angle θ made with the positive x -axis measured counterclockwise. This gives the **polar coordinates** (r, θ) of P .

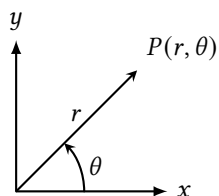


Figure 2.1: Polar coordinates.

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

2.2 Cylindrical coordinates

Given a point P with Cartesian coordinates (x, y, z) in $\mathbb{R}^3$, we may use the polar coordinates (r, θ) for the position of the foot of the perpendicular from P onto the xy -plane. Then the triple (r, θ, z) determines the position of P ; it is called the **cylindrical coordinates** of P .

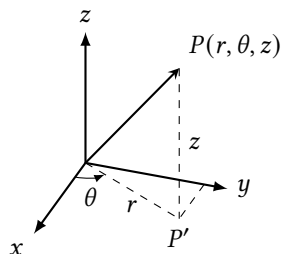


Figure 2.2: Cylindrical coordinates.

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}, \quad z = z$$

2.3 Spherical coordinates

Given a point $P(x, y, z)$, let P' be the foot of the perpendicular from P onto the xy -plane.

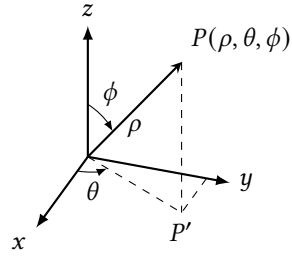


Figure 2.3: Spherical coordinates.

Let $\rho \geq 0$ be its distance from the origin, θ be the angle that OP' makes with the x -axis (measured counterclockwise), and ϕ be the angle that OP makes with the z -axis.

Note that $OP' = \rho \sin \phi$ and $PP' = \rho \cos \phi$.

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad \cos \phi = \frac{z}{\rho}, \quad \cos \theta = \frac{x}{\rho \sin \phi}$$

3 Vector Functions

3.1 Vector functions and space curves

Definition 3.1. A **vector function** $\mathbf{r}(t)$ is a function whose domain is a set of real numbers, whose range is a set of vectors:

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

We call f, g, h the **component functions** of $\mathbf{r}$.

We say that $\mathbf{r}(t)$ is a **parametrization** of C , and t is a **parameter**. Note that the parametrization of a curve is not unique; for example, both

$$\mathbf{r}(t) = \langle t, t^2 \rangle, \quad t \in \mathbb{R}$$

$$\mathbf{r}(t) = \langle t^3, t^6 \rangle, \quad t \in \mathbb{R}$$

parametrize the same parabola $f(x) = x^2$ on the xy -plane, since $t \mapsto t^3$ is a bijection. However, $\mathbf{r}(t) = \langle t, t^2 \rangle$ does not parametrize the same curve, since $t \mapsto t^2$ is not a bijection.

Let $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ be a vector function. Then $\mathbf{r}(t)$ is the position vector of the point $(f(t), g(t), h(t))$ on C . Hence, any continuous vector function $\mathbf{r}$ defines a space curve C traced out by the tip of the moving vector $\mathbf{r}(t)$.

Definition 3.2. Let $\mathbf{r} = \langle f(t), g(t), h(t) \rangle$. The **limit** of $\mathbf{r}(t)$ as $t \rightarrow a$ is

$$\lim_{t \rightarrow a} \mathbf{r}(t) := \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle.$$

That is, the limit of a vector function $\mathbf{r}$ is obtained by taking the limits of its component functions.

Definition 3.3. A vector function $\mathbf{r}(t)$ is **continuous** at $t = a$ if $\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$.

That is, $\mathbf{r}$ is continuous at a if its component functions are continuous at a .

3.2 Derivatives and integrals

Definition 3.4. The **derivative** of a vector function $\mathbf{r}(t)$ at $t = a$ is

$$\mathbf{r}'(a) := \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(a + \Delta t) - \mathbf{r}(a)}{\Delta t}$$

if the limit exists. We also call $\mathbf{r}'(a)$ a **tangent vector** to the curve at $t = a$.

Definition 3.5. The **derivative** of a vector function $\mathbf{r}(t)$ is

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) := \lim_{h \rightarrow 0} \frac{\mathbf{r}(t + h) - \mathbf{r}(t)}{h}$$

if the limit exists.

To find the derivative of a vector function $\mathbf{r}$, we seldom compute from the definition. It turns out that we can simply differentiate component-wise:

Theorem 3.6. Let $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, where f, g, h are differentiable at $t = a$. Then $\mathbf{r}$ is differentiable at $t = a$, and

$$\mathbf{r}'(a) = \langle f'(a), g'(a), h'(a) \rangle. \quad (7)$$

Proof. Write $\mathbf{r}$ in component functions. □

Theorem 3.7 (Differentiation rules). Let $\mathbf{u}(t)$ and $\mathbf{v}(t)$ be differentiable vector functions, c be a scalar, and f a real-valued function. Then

(i) *Addition:* $\frac{d}{dt}(\mathbf{u}(t) + \mathbf{v}(t)) = \mathbf{u}'(t) + \mathbf{v}'(t)$.

(ii) *Scalar multiplication:* $\frac{d}{dt}(c\mathbf{u}(t)) = c\mathbf{u}'(t)$.

(iii) *Product by scalar function:* $\frac{d}{dt}(f(t)\mathbf{u}(t)) = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$.

(iv) *Dot product:* $\frac{d}{dt}(\mathbf{u}(t) \cdot \mathbf{v}(t)) = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$.

(v) *Vector product:* $\frac{d}{dt}(\mathbf{u}(t) \times \mathbf{v}(t)) = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$.

(vi) *Chain rule:* $\frac{d}{dt}(\mathbf{u}(f(t))) = f'(t)\mathbf{u}'(f(t))$.

Corollary 3.8. If $\|\mathbf{r}(t)\| = c$ where c is a constant, then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for all t .

Proof. By hypothesis, $\mathbf{r}(t) \cdot \mathbf{r}(t) = \|\mathbf{r}(t)\|^2 = c^2$ is a constant. By (iv) above,

$$0 = \frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{r}(t)] = \mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 2\mathbf{r}'(t) \cdot \mathbf{r}(t).$$

Hence $\mathbf{r}'(t) \cdot \mathbf{r}(t) = 0$. □

Geometrically, if a curve lies on a sphere with center the origin, then the tangent vector $\mathbf{r}'(t)$ is always perpendicular to the position vector $\mathbf{r}(t)$.

Definition 3.9. Let $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ be continuous. The **definite integral** of $\mathbf{r}(t)$ from $t = a$ to $t = b$ is

$$\int_a^b \mathbf{r}(t) dt := \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle.$$

This means that we can evaluate an integral of a vector function by integrating each component function.

Theorem 3.10 (Fundamental theorem of calculus). Let $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ be continuous. Then

$$\int_a^b \mathbf{r}(t) dt = \mathbf{R}(b) - \mathbf{R}(a) \tag{8}$$

where $\mathbf{R}$ is an anti-derivative of $\mathbf{r}$, i.e., $\mathbf{R}'(t) = \mathbf{r}(t)$.

3.3 Arc length

A natural question about a curve in space is “How long is it?”. This is the **arc length** of the curve. Our formula for arc length only applies to *smooth* curves, i.e., curves that do not have sharp corners at their interior points.

Theorem 3.11 (Arc length formula). Let C be the curve given by $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, $t \in [a, b]$, where f' , g' and h' are continuous. If C is traversed exactly once as t increases from a to b , then the length of C is

$$\begin{aligned} s &= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt \\ &= \int_a^b \|\mathbf{r}'(t)\| dt. \end{aligned} \tag{9}$$

Proof. We can approximate the arc length of this curve as follows:

1. Partition the interval $[a, b]$ into n subintervals of equal size: $a = t_0 < t_1 < \dots < t_n = b$, where $\Delta t = t_i - t_{i-1} = \frac{b-a}{n}$ for all $i = 1, \dots, n$.
2. Let s_i denote the arc length of the portion of the curve traced out as t increases from t_{i-1} to t_i . We can approximate s_i by the distance between the points $(f(t_{i-1}), g(t_{i-1}), h(t_{i-1}))$ and $(f(t_i), g(t_i), h(t_i))$:

$$s_i \approx \sqrt{[f(t_i) - f(t_{i-1})]^2 + [g(t_i) - g(t_{i-1})]^2 + [h(t_i) - h(t_{i-1})]^2}.$$

3. By the mean value theorem, we have

$$\begin{aligned}f(t_i) - f(t_{i-1}) &= f'(c_i)(t_i - t_{i-1}) = f'(c_i)\Delta t \\g(t_i) - g(t_{i-1}) &= g'(d_i)(t_i - t_{i-1}) = g'(d_i)\Delta t \\h(t_i) - h(t_{i-1}) &= h'(e_i)(t_i - t_{i-1}) = h'(e_i)\Delta t\end{aligned}$$

for some points $c_i, d_i, e_i \in (t_{i-1}, t_i)$. This yields

$$s_i \approx \sqrt{[f'(c_i)]^2 + [g'(d_i)]^2 + [h'(e_i)]^2} \Delta t.$$

Notice that when Δt is small, c_i, d_i and e_i are very close to each other. Thus we can further approximate s_i as

$$s_i \approx \sqrt{[f'(c_i)]^2 + [g'(c_i)]^2 + [h'(c_i)]^2} \Delta t.$$

4. The total arc length is then approximately the Riemann sum

$$s \approx \sum_{i=1}^n s_i = \sum_{i=1}^n \sqrt{[f'(c_i)]^2 + [g'(c_i)]^2 + [h'(c_i)]^2} \Delta t$$

where the total error in the approximation tends to 0 as $\Delta t \rightarrow 0$.

5. We can make $\Delta t \rightarrow 0$ by taking $n \rightarrow \infty$. This gives exact length:

$$s = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{[f'(c_i)]^2 + [g'(c_i)]^2 + [h'(c_i)]^2} \Delta t$$

provided the limit exists.

□

4 Partial Derivatives

4.1 Functions of several variables

Definition 4.1. A **function** $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ of two variables is a rule that assigns to each ordered pair of real numbers $(x, y) \in D$ a unique real number, denoted by $f(x, y)$.

The **domain** of f is the set D . If no domain is specified, then the domain of f is understood to be the set of all pairs (x, y) for which $f(x, y)$ is a well-defined real number.

The **range** of f is the set of values that f takes on:

$$\{f(x, y) : (x, y) \in D\}.$$

We write $z = f(x, y)$ to indicate that z is a function of x and y . We call x and y the **independent variables**, and z the **dependent variable**.

The graph of a function f of two variables is also called the **surface** S with equation $z = f(x, y)$. We can visualize the graph S of f as lying directly above or below its domain D in the xy -plane.

[figure]

Definition 4.2. Let f be a function of two variables.

- **Horizontal traces (level curves):** resulting curves when we intersect the surface $z = f(x, y)$ with horizontal planes, parallel to xy -plane.
- **Vertical traces:** resulting curves when we intersect the surface $z = f(x, y)$ with vertical planes.

To visualise the graph for $f(x, y)$, we can take hints from its horizontal and vertical traces.

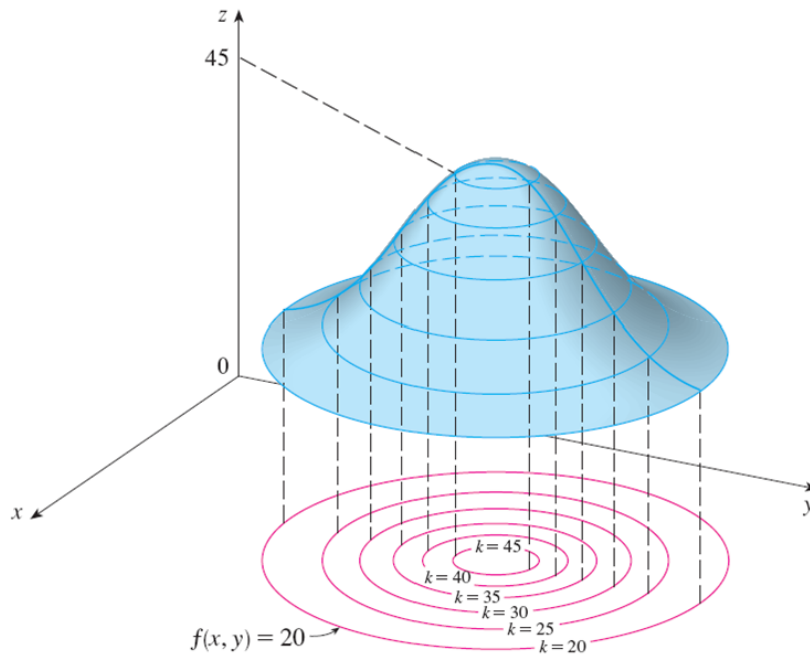
Definition 4.3. A **level curve** of $f(x, y)$ is the two-dimensional graph of the equation $f(x, y) = k$ for some constant k .

A level curve is a curve in the xy -plane that shows where the graph of f has height k (above or below the xy -plane). It is the intersection of the surface and the plane $z = k$.

Definition 4.4. A **contour plot** of $f(x, y)$ is a graph of numerous level curves $f(x, y) = k$, for representative values of k .

To sketch contour plots, we use values of k that are equally spaced. The surface is:

- steep where the level curves are close together.
- flatter where the level curves are farther apart.



Definition 4.5. A function $f: D \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ of three variables is a rule that assigns to each $(x, y, z) \in D$ a unique real number denoted by $f(x, y, z)$.

We can describe f by examining the level surfaces of f :

Definition 4.6. A **level surface** of $f(x, y, z)$ is the three-dimensional graph of the equation $f(x, y, z) = k$ for some constant k .

If the point (x, y, z) moves along a level surface, the value of $f(x, y, z)$ remains fixed.

4.2 Cylinders and quadric surfaces

Definition 4.7. A surface is a cylinder if there is a plane P such that all the planes parallel to P intersect the surface in the same curve (when viewed in 2-dimension).

Definition 4.8. A **quadric surface** is the graph of a second-degree equation in x , y and z :

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

where $A, B, \dots, J$ are constants.

Using translation and rotation (**change of coordinates** after completing squares), the equation of a quadric surface can be expressed in one of two **standard forms**:

$$Ax^2 + By^2 + Cz^2 + J = 0$$

$$Ax^2 + By^2 + Iz = 0.$$

To visualise surfaces, consider horizontal and vertical traces (fix one of x, y, z , then consider the 2D graph).

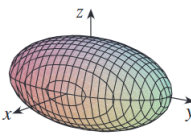
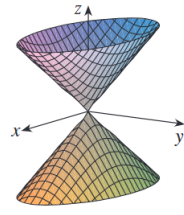
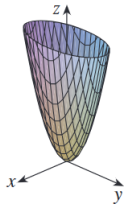
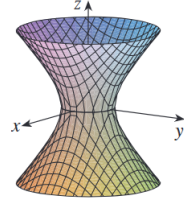
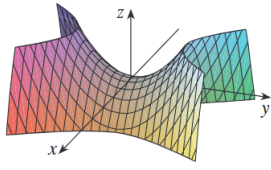
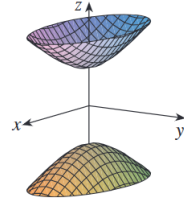
Surface	Equation	Surface	Equation
Ellipsoid 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ All traces are ellipses. If $a = b = c$, the ellipsoid is a sphere.	Cone 	$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces in the planes $x = k$ and $y = k$ are hyperbolas if $k \neq 0$ but are pairs of lines if $k = 0$.
Elliptic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces are parabolas. The variable raised to the first power indicates the axis of the paraboloid.	Hyperboloid of One Sheet 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ Horizontal traces are ellipses. Vertical traces are hyperbolas. The axis of symmetry corresponds to the variable whose coefficient is negative.
Hyperbolic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ Horizontal traces are hyperbolas. Vertical traces are parabolas. The case where $c < 0$ is illustrated.	Hyperboloid of Two Sheets 	$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ Horizontal traces in $z = k$ are ellipses if $k > c$ or $k < -c$. Vertical traces are hyperbolas. The two minus signs indicate two sheets.

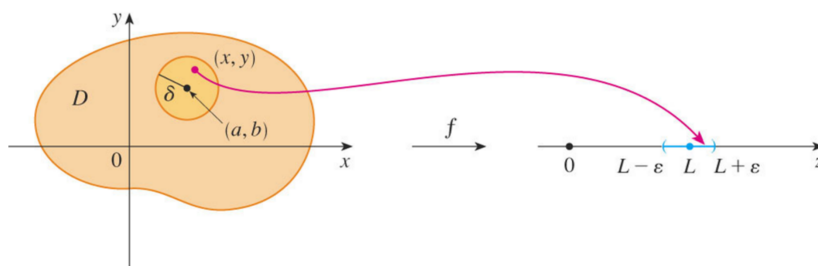
Figure 4.1: Graphs of quadric surfaces.

4.3 Limits and continuity

Definition 4.9. Let $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. We say that the **limit** of f as (x, y) approaches (a, b) is L , and write $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$, if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall (x, y) \in D, \quad 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x, y) - L| < \varepsilon.$$

In other words, we can make the values of $f(x, y)$ as close to L as we like by taking the point (x, y) sufficiently close to the point (a, b) , but not equal to (a, b) .



Lemma 4.10. If $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$ exists, then the limit L is unique.

Proof. Suppose $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L_1$ and $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L_2$. Let $\varepsilon > 0$ be given. There exist $\delta_1, \delta_2 > 0$ such that for all $(x, y) \in D$,

$$\begin{aligned} 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta_1 &\implies |f(x, y) - L_1| < \frac{\varepsilon}{2}, \\ 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta_2 &\implies |f(x, y) - L_2| < \frac{\varepsilon}{2}. \end{aligned}$$

Take $\delta = \min\{\delta_1, \delta_2\}$. If $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$, then

$$|L_1 - L_2| \leq |L_1 - f(x, y)| + |f(x, y) - L_2| < \varepsilon.$$

Since this holds for arbitrary ε , we conclude that $L_1 = L_2$. $\square$

The next result states that if a function $f(x, y)$ approaches a limit L as $(x, y) \rightarrow (a, b)$, then the limit of f along any path that approaches (a, b) must also be L .

Theorem 4.11. Let $x = \alpha(t)$, $y = \beta(t)$ be the parametric equations of a path in $\mathbb{R}^2$ such that $(\alpha(t), \beta(t))$ lies in the domain of $f(x, y)$ for all t in a certain open interval containing t_0 and $\lim_{t \rightarrow t_0} \alpha(t) = a$ and $\lim_{t \rightarrow t_0} \beta(t) = b$. If $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$, then $\lim_{t \rightarrow t_0} f(\alpha(t), \beta(t)) = L$.

Proof. Let $\varepsilon > 0$ be given. There exists $\delta > 0$ such that for all $(x, y) \in D$,

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x, y) - L| < \varepsilon.$$

Since $\lim_{t \rightarrow t_0} \alpha(t) = a$ and $\lim_{t \rightarrow t_0} \beta(t) = b$, there exists $\eta > 0$ such that

$$0 < |t - t_0| < \eta \implies |\alpha(t) - a| < \frac{\delta}{2} \quad \text{and} \quad |\beta(t) - b| < \frac{\delta}{2}.$$

Thus for all t satisfying $0 < |t - t_0| < \eta$, we have

$$0 < \sqrt{[\alpha(t) - a]^2 + [\beta(t) - b]^2} < \sqrt{\frac{\delta^2}{4} + \frac{\delta^2}{4}} < \delta$$

so that

$$|f(\alpha(t), \beta(t)) - L| < \varepsilon.$$

This shows that $\lim_{t \rightarrow t_0} f(\alpha(t), \beta(t)) = L$. $\square$

By contrapositive, if the function has different limits along different paths of approach, then the limit does not exist. If we suspect the limit of f does not exist at (a, b) , some paths to try are:

- $x = a, y \rightarrow b$ (vertical lines);
- $y = b, x \rightarrow a$ (horizontal lines);
- $y = g(x), x \rightarrow a$ where $g(x)$ is some simple function such that $g(a) = b$;
- $x = g(y), y \rightarrow b$ where $g(y)$ is some simple function such that $g(b) = a$.

The usual limit theorems hold for limits of functions of two variables:

- (i) The limit of a sum/difference is the sum/difference of the limits.
- (ii) The limit of a scalar multiple of a function is the scalar multiple of the limit of the function.

(iii) The limit of a product is the product of the limits.

(iv) The limit of a quotient is the quotient of the limits (provided that the limit of the denominator is non-zero).

Theorem 4.12 (Squeeze theorem). Suppose that $|f(x, y) - L| \leq g(x, y)$ for all (x, y) in the interior of some circle centered at (a, b) , except possibly at (a, b) . If $\lim_{(x,y) \rightarrow (a,b)} g(x, y) = 0$, then

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L.$$

Proof. exercise, proof using epsilon delta □

Definition 4.13. Let $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. We say that f is **continuous** at (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

We say that f is **continuous** on D if f is continuous at every point in D .

Theorem 4.14. Let f and g be continuous at (a, b) . Then

$$f \pm g, \quad cf, \quad fg, \quad \frac{f}{g}$$

are all continuous at (a, b) , provided $g(a, b) \neq 0$.

Proof. Use the limit laws. □

Subsequently, the following classes of functions are continuous in their domains:

- Polynomials
- Trigonometric and exponential functions
- Rational functions

Theorem 4.15. If f is continuous at (a, b) and g is continuous at $f(a, b)$, then $g \circ f$ is continuous at (a, b) .

One may compute limits using **polar coordinates**. This is especially convenient for limits at the origin and for those expressions that are independent of θ .

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{r \rightarrow 0^+} f(r \cos \theta, r \sin \theta).$$

4.4 Partial derivatives

Recall that for a function f of a single variable, we defined the derivative function as

$$f'(x) := \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

To extend this to multivariable functions, the idea is to vary one variable and keep other variable(s) fixed.

Definition 4.16. Let f be a function of two variables. The **partial derivatives** of f are

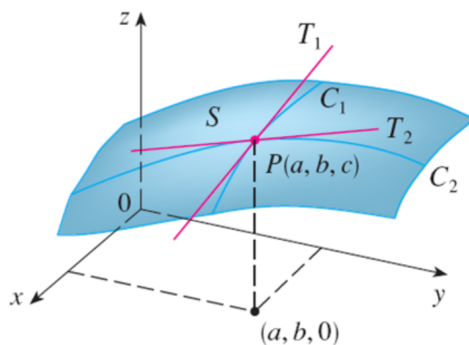
$$f_x(x, y) := \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h},$$

$$f_y(x, y) := \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}.$$

We also denote $f_x = \frac{\partial f}{\partial x}$ and $f_y = \frac{\partial f}{\partial y}$.

To compute the partial derivative of a function $f(x, y)$ or $f(x, y, z)$ or $f(x_1, \dots, x_n)$ with respect to one of its input variables, treat all other variables as constants, and differentiate f with respect to that variable.

Consider the geometric interpretation of partial derivatives. Let S be the surface given by the equation $z = f(x, y)$.



Fix $y = b$; the curve C_1 is the graph of the function $g(x) = f(x, b)$. By definition, $f_x(a, b)$ equals the gradient of the tangent line T_1 , i.e., the rate of change of f along the x -direction at the point P :

$$g'(a) := \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h} =: f_x(a, b).$$

Similarly, fix $x = a$. The curve C_2 is the graph of the function $h(y) = f(a, y)$. Then $f_y(a, b)$ equals the gradient of the tangent line T_2 , i.e., the rate of change of f along the y -direction at the point P : $h'(b) = f_y(a, b)$.

For functions of more than two variables $w = f(x, y, z)$, similarly we define the partial derivatives

$$\begin{aligned} f_x &= \frac{\partial f}{\partial x} = \frac{\partial w}{\partial x} \\ f_y &= \frac{\partial f}{\partial y} = \frac{\partial w}{\partial y} \\ f_z &= \frac{\partial f}{\partial z} = \frac{\partial w}{\partial z} \end{aligned}$$

For **second-order partial derivatives**, we write

$$\begin{aligned} f_{xx} &= (f_x)_x = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} \\ f_{xy} &= (f_x)_y = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} \\ f_{yx} &= (f_y)_x = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} \\ f_{yy} &= (f_y)_y = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} \end{aligned}$$

Theorem 4.17 (Clairaut's theorem). *Let f be defined on an open disk D that contains the point (a, b) . If f_{xy} and f_{yx} are continuous at (a, b) , then*

$$f_{xy}(a, b) = f_{yx}(a, b). \quad (10)$$

Proof. Let f_{xy} and f_{yx} be defined on an open disk D centered at (a, b) . Let $(x, y) \in D$. Fix x consider the following function in y :

$$[f(x, y) - f(a, y)] - [f(x, b) - f(a, b)].$$

Applying the mean value theorem with respect to y , there exists some ξ_1 between y and b such that

$$[f(x, y) - f(a, y)] - [f(x, b) - f(a, b)] = [f_y(x, \xi_1) - f_y(a, \xi_1)](y - b)$$

Next, applying mean value theorem to $f_y(x, \xi_1)$ with respect to x , there exists some ξ_2 between x and a such that

$$[f(x, y) - f(a, y)] - [f(x, b) - f(a, b)] = f_{yx}(\xi_2, \xi_1)(x - a)(y - b).$$

Now we can rewrite the expression

$$[f(x, y) - f(a, y)] - [f(x, b) - f(a, b)] = [f(x, y) - f(x, b)] - [f(a, y) - f(a, b)].$$

Applying mean value theorem first with respect to x and then with respect to y , there exists ξ_3 between x and a , ξ_4 between y and b , such that

$$[f(x, y) - f(a, y)] - [f(x, b) - f(a, b)] = [f(x, y) - f(x, b)] - [f(a, y) - f(a, b)] = f_{xy}(\xi_3, \xi_4)(y - b)(x - a).$$

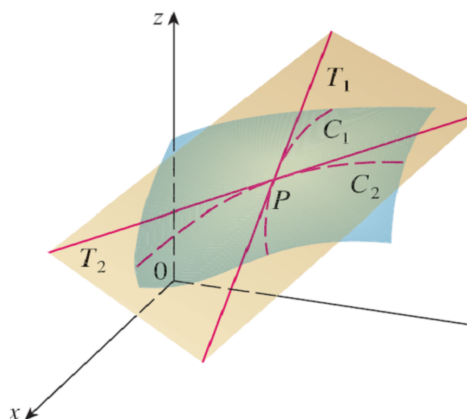
Thus we have $f_{xy}(\xi_2, \xi_1) = f_{yx}(\xi_3, \xi_4)$.

Since f_{xy} and f_{yx} are continuous at (a, b) , by taking the limit $(x, y) \rightarrow (a, b)$, we obtain $f_{xy}(a, b) = f_{yx}(a, b)$. $\square$

4.5 Tangent planes

Recall that in single-variable calculus, as we zoom in toward a point on the graph of a differentiable function, the graph becomes indistinguishable from its tangent line. Thus, we can approximate the function by a linear function.

Let f be a function of two variables. The graph of f is a surface in $\mathbb{R}^3$ with equation $z = f(x, y)$. Let $P(a, b, c)$ be a point on the surface; thus $c = f(a, b)$.



The **tangent plane** to the surface S at the point P is defined to be the plane that contains both tangent lines T_1 and T_2 .

Since P lies on the tangent plane, it suffices to find a normal vector $\mathbf{n}$. Recall that $f_x(a, b)$ is the gradient of tangent line T_1 . Thus, a vector with the same direction as T_1 is

$$\langle 1, 0, f_x(a, b) \rangle.$$

Similarly, $f_y(a, b)$ is the gradient of tangent line T_2 , so a vector with the same direction as T_2 is

$$\langle 0, 1, f_y(a, b) \rangle.$$

Hence, a vector normal to the plane is the cross product

$$\mathbf{n} = \langle 0, 1, f_y(a, b) \rangle \times \langle 1, 0, f_x(a, b) \rangle = \langle f_x(a, b), f_y(a, b), -1 \rangle.$$

Theorem 4.18. The equation of the tangent plane of f at (a, b) is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b). \quad (11)$$

Recall that for a function of one variable $y = f(x)$, if x changes from a to $a + \Delta x$, the increment of y is

$$\Delta y = f(a + \Delta x) - f(a).$$

If f is differentiable at a , then

$$\Delta y = f'(a)\Delta x + \varepsilon\Delta x$$

where $\varepsilon \rightarrow 0$ as $\Delta x \rightarrow 0$.

Definition 4.19. Let $z = f(x, y)$. Suppose Δx and Δy are increments in the independent variables x and y respectively from (a, b) . Then the **increment** in z at (a, b) is

$$\Delta z := f(a + \Delta x, b + \Delta y) - f(a, b).$$

Definition 4.20. Let $z = f(x, y)$. We say that f is **differentiable** at (a, b) if

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \underbrace{\varepsilon_1\Delta x + \varepsilon_2\Delta y}_{\text{error terms}}$$

where $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$. We say that f is **differentiable** on a region $R \subseteq \mathbb{R}^2$ if f is differentiable at every point in R .

From the definition of differentiability, if $f(x, y)$ is differentiable at (a, b) , then $f_x(a, b)$ and $f_y(a, b)$ exist. However, the converse is not true.

Lemma 4.21. If f is differentiable at (a, b) , then f is continuous at (a, b) .

It is sometimes hard to directly use the definition to check the differentiability of a function, but the next theorem provides a convenient sufficient condition for differentiability.

Theorem 4.22. If $f_x(x, y)$ and $f_y(x, y)$ exist in an open disk containing (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .

Proof. Write

$$\begin{aligned}\Delta z &= f(a + \Delta x, b + \Delta y) - f(a, b) \\ &= [f(a + \Delta x, b + \Delta y) - f(a + \Delta x, b)] + [f(a + \Delta x, b) - f(a, b)].\end{aligned}$$

By assumption, f_x and f_y exist near (a, b) . Thus when Δx and Δy are sufficiently small, we may apply the Mean Value Theorem to each of the above differences. Then

$$\Delta z = f_y(a + \Delta x, b + c_1\Delta y)\Delta y + f_x(a + c_2\Delta x, b)\Delta x$$

where $c_1, c_2 \in (0, 1)$. Moreover, by the assumption that both f_x and f_y are continuous at (a, b) , we have

$$\begin{aligned}f_y(a + \Delta x, b + c_1\Delta y) &= f_y(a, b) + \varepsilon_2, \\ f_x(a + c_2\Delta x, b) &= f_x(a, b) + \varepsilon_1,\end{aligned}$$

where $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$. Thus

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

where $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$. Therefore f is differentiable at (a, b) . □

From Definition 4.20, notice that

$$\underbrace{\Delta z}_{\text{change in the function at } (a, b)} \approx \underbrace{f_x(a, b)\Delta x + f_y(a, b)\Delta y}_{\text{change in tangent plane}}$$

since the error terms tend to 0. Hence, adding $f(a, b)$ on both sides gives

$$f(a + \Delta x, b + \Delta y) \approx f(a, b) + f_x(a, b)\Delta x + f_y(a, b)\Delta y.$$

As $\Delta x, \Delta y \rightarrow 0$, the tangent plane gets closer and closer to the surface. Thus, we may use the function defining the tangent plane as an approximation of f :

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \quad (12)$$

is called the **linear approximation** of f at (a, b) . Then $f(x, y) \approx L(x, y)$ near (a, b) .

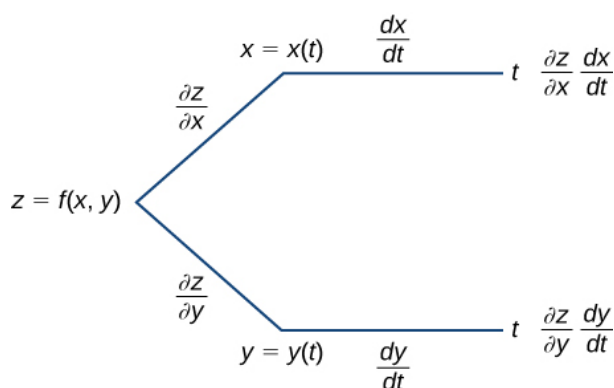
4.6 Chain rule

Recall the chain rule for functions of one variable: If $y = f(x)$ and $x = g(t)$, where f and g are differentiable, then y is a differentiable function of t , and

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}.$$

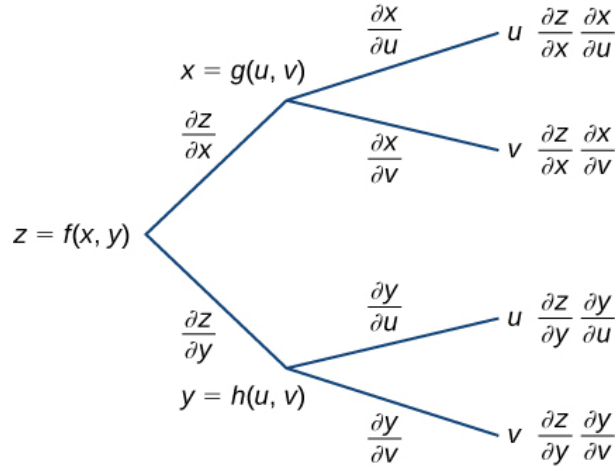
Theorem 4.23 (Chain rule I). *Let $z = f(x, y)$ be a differentiable function of x and y , where $x = g(t)$, $y = h(t)$ are differentiable functions of t . Then z is a differentiable function of t , and*

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}. \quad (13)$$



Theorem 4.24 (Chain rule II). *Let $z = f(x, y)$ be a differentiable function of x and y , where $x = g(s, t)$, $y = h(s, t)$ are differentiable functions of s and t . Then z is a differentiable function of s and t , and*

$$\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, \\ \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}. \end{aligned} \quad (14)$$



Theorem 4.25 (Chain rule, general). Suppose that u is a differentiable function of n variables $x_1, \dots, x_n$, and each x_j is a differentiable function of m variables $t_1, \dots, t_m$. Then u is a function of $t_1, \dots, t_m$ and

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i} \quad (15)$$

for each $i = 1, \dots, m$.

4.7 Implicit differentiation

Recall that implicit differentiation provides a method for finding $\frac{dy}{dx}$ when y is defined implicitly as a function of x . The method involves differentiating both sides of the equation defining the function with respect to x , then solving for $\frac{dy}{dx}$.

Suppose $F(x, y) = 0$ defines y implicitly as a function of x . Use the chain rule to differentiate F with respect to x :

$$\frac{\partial F}{\partial x} \frac{dx}{dx} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0.$$

But $\frac{dx}{dx} = 1$, so if $\frac{\partial F}{\partial y} \neq 0$, we have

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{F_x}{F_y}. \quad (16)$$

Suppose $F(x, y, z) = 0$ defines z implicitly as a function of x and y . Use the chain rule to differentiate $F(x, y, z) = 0$ with respect to x :

$$\frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0.$$

But $\frac{\partial x}{\partial x} = 1$ and $\frac{\partial y}{\partial x} = 0$, so $\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0$. If $\frac{\partial F}{\partial z} \neq 0$, we solve for $\frac{\partial z}{\partial x}$:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}. \quad (17)$$

4.8 The gradient

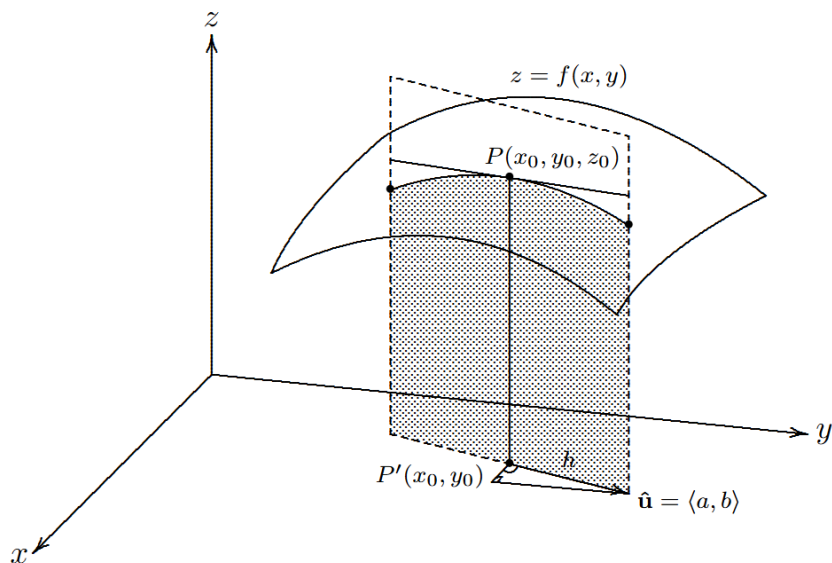
Definition 4.26. The **directional derivative** of $f(x, y)$ at (x_0, y_0) in the direction of a unit vector $\mathbf{u} = \langle a, b \rangle$ is

$$D_{\mathbf{u}}f(x_0, y_0) := \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

if the limit exists.

Geometrically, the directional derivative $D_{\mathbf{u}}f(x_0, y_0)$ is the rate of change of the function at the point (x_0, y_0) in

the direction given by $\mathbf{u}$:



In particular, $D_i f(x_0, y_0) = f_x(x_0, y_0)$ and $D_j f(x_0, y_0) = f_y(x_0, y_0)$. Thus, the partial derivatives of f with respect to x and y are special cases of the directional derivative.

Theorem 4.27. *Let $f(x, y)$ be differentiable. Then f has a directional derivative in the direction of any unit vector $\mathbf{u} = \langle a, b \rangle$, and*

$$\begin{aligned} D_{\mathbf{u}} f(x, y) &= f_x(x, y)a + f_y(x, y)b \\ &= \langle f_x(x, y), f_y(x, y) \rangle \cdot \mathbf{u}. \end{aligned} \tag{18}$$

Proof. Define a function g of the single variable h by

$$g(h) = f(x_0 + ha, y_0 + hb).$$

On one hand, by definition of the derivative, we have

$$\begin{aligned} g'(0) &= \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h} \\ &= D_{\mathbf{u}} f(x_0, y_0). \end{aligned}$$

On the other hand, by the chain rule,

$$g'(h) = \frac{\partial f}{\partial x} \frac{dx}{dh} + \frac{\partial f}{\partial y} \frac{dy}{dh} = f_x(x, y)a + f_y(x, y)b.$$

At $h = 0$, then $x = x_0$, $y = y_0$, so

$$g'(0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b.$$

This yields the desired result. □

If the unit vector $\mathbf{u}$ makes an angle θ with the positive x -axis, write $\mathbf{u} = \langle \cos \theta, \sin \theta \rangle$. By (18),

$$D_{\mathbf{u}} f(x, y) = f_x(x, y) \cos \theta + f_y(x, y) \sin \theta.$$

Definition 4.28. The **gradient** of $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, denoted ∇f , is the vector-valued function

$$\nabla f(x, y) := \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j},$$

provided both partial derivatives exist.

By (18), if $\mathbf{u}$ is a unit vector, then

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}. \quad (19)$$

We proceed similarly for functions of three variables.

Definition 4.29. The **directional derivative** of $f(x, y, z)$ at (x_0, y_0, z_0) in the direction of a unit vector $\mathbf{u} = \langle a, b, c \rangle$ is

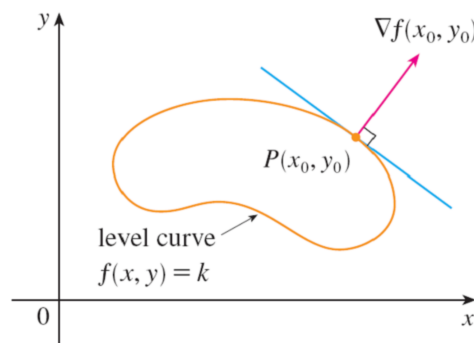
$$\begin{aligned} D_{\mathbf{u}}f(x_0, y_0, z_0) &:= \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb, z_0 + hc) - f(x_0, y_0, z_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(\mathbf{x}_0 + h\mathbf{u}) - f(\mathbf{x}_0)}{h} \end{aligned}$$

if the limit exists. Similarly, the **gradient** of a differentiable function f is

$$\nabla f := \langle f_x, f_y, f_z \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}.$$

Then we have the formula $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

Theorem 4.30. Let $f(x, y)$ be differentiable at (x_0, y_0) . Suppose $\nabla f(x_0, y_0) \neq \mathbf{0}$. Then $\nabla f(x_0, y_0)$ is normal to the level curve $f(x, y) = k$ that contains the point (x_0, y_0) .



Proof. Let $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ be a parametrisation of the level curve $f(x, y) = k$. Now differentiate both sides of the equation $f(x, y) = k$ with respect to t , using the chain rule for the LHS:

$$\begin{aligned} f_x x'(t) + f_y y'(t) &= 0 \\ \langle f_x, f_y \rangle \cdot \langle x'(t), y'(t) \rangle &= 0 \\ \langle f_x, f_y \rangle \cdot \mathbf{r}'(t) &= 0 \end{aligned}$$

Since the tangent vector $\mathbf{r}'(t)$ is perpendicular to the level curve, and ∇f is perpendicular to $\mathbf{r}'(t)$, this means that ∇f is normal to the level curve. $\square$

The next result tells us that f has the maximum rate of change in the direction of the gradient vector ∇f .

Theorem 4.31. Let f be a differentiable function of two or three variables. Then

$$D_{\mathbf{u}}f(\mathbf{x}) \leq \|\nabla f(\mathbf{x})\| \quad (20)$$

where equality holds if and only if $\mathbf{u}$ has the same direction as the gradient vector $\nabla f(\mathbf{x})$.

Proof. By (18),

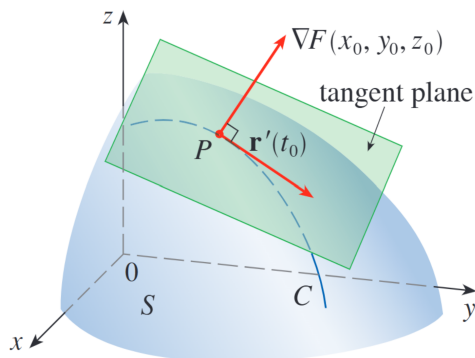
$$D_{\mathbf{u}}f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{u} = \|\nabla f(\mathbf{x})\| \|\mathbf{u}\| \cos \theta = \|\nabla f(\mathbf{x})\| \cos \theta.$$

Since $\cos \theta \leq 1$, this gives the desired inequality.

Equality holds if and only if $\cos \theta = 1$, i.e., $\theta = 0$, when $\mathbf{u}$ has the same direction as $\nabla f(\mathbf{x})$. $\square$

4.9 Tangent planes to level surfaces

Let S be a surface with equation $F(x, y, z) = k$, where k is a constant; that is, S is a level surface of F . Let $P(x_0, y_0, z_0)$ be a point on S . We want to find the equation of the tangent plane to S at P .



Take any curve C on the surface S , described by $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$. Let t_0 be the parameter value corresponding to P ; that is, $\mathbf{r}(t_0) = \langle x_0, y_0, z_0 \rangle$. Since C lies on S , any point $(x(t), y(t), z(t))$ satisfies the equation of S :

$$F(x(t), y(t), z(t)) = k.$$

Use the chain rule to differentiate both sides with respect to t :

$$\frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} + \frac{\partial F}{\partial z} \frac{dz}{dt} = 0.$$

But since $\nabla F = \langle F_x, F_y, F_z \rangle$ and $\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$, we rewrite

$$\nabla F \cdot \mathbf{r}'(t) = 0.$$

At $t = t_0$, we have $\nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0$. This means that the gradient vector at P is perpendicular to the tangent vector to any curve C on S that passes through P .

Theorem 4.32. Let $F(x, y, z)$ be differentiable at (x_0, y_0, z_0) . Suppose $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$. Then $\nabla F(x_0, y_0, z_0)$ is normal to the level surface $F(x, y, z) = k$ that contains the point (x_0, y_0, z_0) .

Consequently, the **tangent plane** to the level surface $F(x, y, z) = k$ at $P(x_0, y_0, z_0)$ is the plane that passes through P and has normal vector $\nabla F(x_0, y_0, z_0)$:

$$\langle x - x_0, y - y_0, z - z_0 \rangle \cdot \nabla F(x_0, y_0, z_0) = 0 \quad (21)$$

or

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

In the special case where the surface S is the graph of a function $z = f(x, y)$, we can regard S as the level surface

$$F(x, y, z) = f(x, y) - z = 0.$$

In other words, the graph of $z = f(x, y)$ is simply the level surface of $F(x, y, z)$ at level 0. In this case, $\nabla F = \langle f_x, f_y, -1 \rangle$ is a normal vector to the tangent plane of S at $(x, y, f(x, y))$.

Therefore, if we consider a point $P(x_0, y_0, f(x_0, y_0))$ on the graph of $z = f(x, y)$, the equation of the tangent plane is

$$\langle x - x_0, y - y_0, z - f(x_0, y_0) \rangle \cdot \langle f_x, f_y, -1 \rangle = 0.$$

That is,

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

This is the same formula obtained in

4.10 Maximum and minimum values

Definition 4.33. Let $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. Then

- f has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ for all points in some disk with center (a, b) . The number $f(a, b)$ is called a **local maximum value**.
- f has an **absolute maximum** at (a, b) if $f(x, y) \leq f(a, b)$ for all points in the domain D . The number $f(a, b)$ is called an **absolute maximum value**.

Definition 4.34. Let $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. Then

- f has a **local minimum** at (a, b) if $f(x, y) \geq f(a, b)$ for all points in some disk with center (a, b) . The number $f(a, b)$ is called a **local minimum value**.
- f has an **absolute minimum** at (a, b) if $f(x, y) \geq f(a, b)$ for all points in the domain D . The number $f(a, b)$ is called an **absolute minimum value**.

Theorem 4.35. If f has a local maximum or minimum at (a, b) , and $f_x(a, b)$ and $f_y(a, b)$ exist, then

$$f_x(a, b) = f_y(a, b) = 0.$$

That is, $\nabla f(a, b) = 0$.

Proof. Let $g(x) = f(x, b)$, where we keep $y = b$ fixed. If f has a local maximum [minimum] at (a, b) , then g has a local maximum [minimum] at a . By Fermat's theorem, $g'(a) = 0$. But $g'(a) = f_x(a, b)$. Thus $f_x(a, b) = 0$.

Similarly, by applying Fermat's theorem to $h(y) = f(a, y)$, we obtain $f_y(a, b) = 0$. $\square$

Substituting $f_x(a, b) = 0$ and $f_y(a, b) = 0$ into the equation of tangent plane (11), we get $z = f(a, b)$. Hence, the tangent plane at a local maximum or minimum is a **horizontal plane** parallel to the xy -plane.

Definition 4.36. Let $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. A point (a, b) is called a **critical point** of f if

$$f_x(a, b) = 0 \quad \text{and} \quad f_y(a, b) = 0.$$

If (a, b) is a local maximum/minimum, then (a, b) is a critical point. However, the converse is not true!

Definition 4.37. Let $f: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. A point (a, b) is a **saddle point** of f if

- it is a critical point of f ;
- every open disk centered at (a, b) contains points $(x, y) \in D$ for which $f(x, y) < f(a, b)$, and points $(x, y) \in D$ for which $f(x, y) > f(a, b)$.

In other words, f has a saddle point at (a, b) if there are some directions along which f has a local maximum at (a, b) , and some directions along which f has a local minimum at (a, b) .

Absolute maximum/minimum do not always exist. However, if f is defined on a closed and bounded region, then absolute maximum and minimum always exist.

Definition 4.38. A **closed** set in $\mathbb{R}^2$ is one that contains all its boundary points.

Definition 4.39. A **bounded** set in $\mathbb{R}^2$ is one that is contained in some disk.

Theorem 4.40 (Extreme value theorem). *If f is continuous on a closed and bounded set $D \subseteq \mathbb{R}^2$, then f attains absolute maximum and minimum values in D .*

Closed Interval Method: find absolute maximum and minimum values of a function defined on a closed and bounded set D .

1. Interior: find critical points of f by setting $\nabla f = 0$.
2. Boundary: find extreme values of f on boundary of D (may need to partition the boundary into line segments, or use Lagrange multipliers).
3. The largest of the values is the absolute maximum value.
The smallest of the values is the absolute minimum value.

Theorem 4.41 (Second derivative test). *Suppose f_{xx} , f_{xy} , f_{yx} and f_{yy} are continuous on a disk with center (a, b) , and suppose $f_x(a, b) = 0$, $f_y(a, b) = 0$. Let*

$$D = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

- (i) *If $D > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b) .*
- (ii) *If $D > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b) .*
- (iii) *If $D < 0$, then f has a saddle point at (a, b) .*

If $D = 0$, then no conclusion can be drawn; the point can be a local maximum, a local minimum, a saddle point, or neither.

Remark. To remember the formula of D , consider the **Hessian** of f :

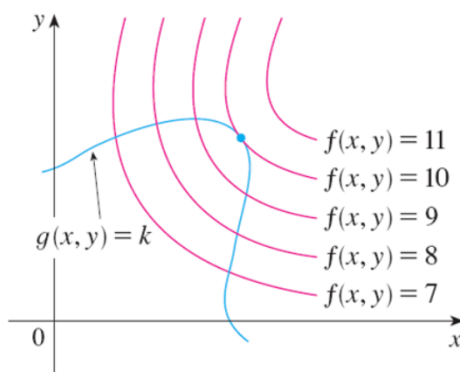
$$H(f) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

Then $D = |H(f)|$.

4.11 Lagrange multipliers

4.11.1 Two variables

In this section, we consider the problem of maximizing or minimizing a function $f(x, y)$ subject to a constraint $g(x, y) = k$. Geometrically, consider the constraint curve $g(x, y) = k$ together with level curves of $z = f(x, y)$.



We see that the maximum value of f occurs at the point (x_0, y_0) where the two curves **just touch each other**, i.e., they have a common tangent line (otherwise, the value of $f(x_0, y_0)$ could be increased further, contradicting the maximality of $f(x_0, y_0)$).

Hence, the gradient vectors are parallel:

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0)$$

for some $\lambda \in \mathbb{R}$, called a **Lagrange multiplier**.

Theorem 4.42 (Lagrange multipliers, two variables). *Let $f(x, y)$ and $g(x, y)$ be differentiable functions such that $\nabla g(x, y) \neq \mathbf{0}$ on the constraint curve $g(x, y) = k$. Suppose that the minimum/maximum value of $f(x, y)$ subject to the constraint $g(x, y) = k$ occurs at (x_0, y_0) . Then*

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0) \quad (22)$$

for some $\lambda \in \mathbb{R}$.

Method of Lagrange Multipliers (two variables case):

1. Find all values of x, y and λ such that $\nabla f(x, y) = \lambda \nabla g(x, y)$ and $g(x, y) = k$.
2. Evaluate f at all the points obtained in Step 1.
 - The largest of these values is the maximum value of f .
 - The smallest is the minimum value of f .

4.11.2 Three variables

Theorem 4.43 (Lagrange multipliers, three variables). *Let $f(x, y, z)$ and $g(x, y, z)$ be differentiable functions such that $\nabla g(x, y, z) \neq \mathbf{0}$ on the constraint curve $g(x, y, z) = k$. Suppose that the minimum/maximum value of $f(x, y, z)$ subject to the constraint $g(x, y, z) = k$ occurs at (x_0, y_0, z_0) . Then*

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0) \quad (23)$$

for some constant λ , called a **Lagrange multiplier**.

Method of Lagrange Multipliers (three variables case):

1. Find all values of x, y, z and λ such that $\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$ and $g(x, y, z) = k$.
2. Evaluate f at all the points obtained in Step 1.
 - The largest of these values is the maximum value of f .
 - The smallest is the minimum value of f .

Writing (23) in terms of components gives a system of equations

$$\begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ f_z = \lambda g_z \\ g(x, y, z) = k \end{cases}$$

4.11.3 More than one constraint

The method of Lagrange Multipliers can be applied to the case of more than one constraint. Consider the problem of maximizing or minimizing $f(x, y, z)$ subject to the constraints $g(x, y, z) = k$ and $h(x, y, z) = c$. If f attains an

extreme value at (x_0, y_0, z_0) , then

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0) + \mu \nabla h(x_0, y_0, z_0) \quad (24)$$

for some λ, μ . In this case, we have **two** Lagrange multipliers. In terms of components, this is equivalent to the following system:

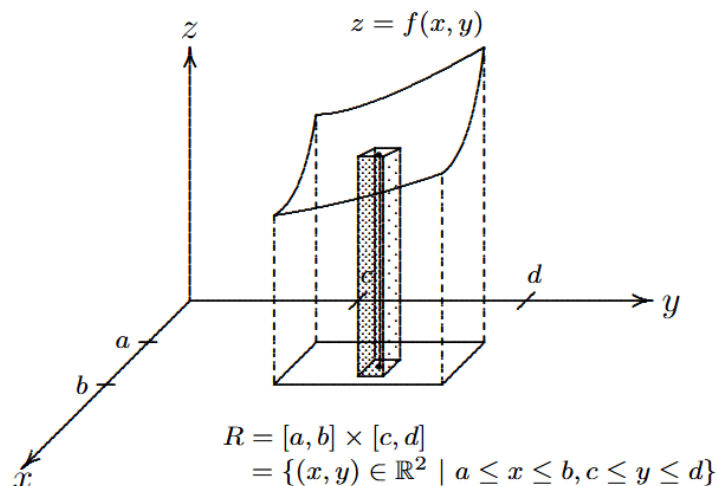
$$\begin{cases} f_x = \lambda g_x + \mu h_x \\ f_y = \lambda g_y + \mu h_y \\ f_z = \lambda g_z + \mu h_z \\ g(x, y, z) = k \\ h(x, y, z) = c \end{cases}$$

5 Multiple Integrals

5.1 Double integrals

5.1.1 Double integrals over rectangles

Let f be a function of two variables defined over a closed rectangle $R = [a, b] \times [c, d]$. We want to define the double integral of f over R as the **volume** of the solid under the graph of $z = f(x, y)$ over R .



1. Divide the rectangle R into sub-rectangles.

- Divide the interval $[a, b]$ into m sub-intervals $[x_{i-1}, x_i]$ of equal length $\Delta x = \frac{b-a}{m}$
- Divide the interval $[c, d]$ into n sub-intervals $[y_{j-1}, y_j]$ of equal length $\Delta y = \frac{d-c}{n}$.

This yields mn sub-rectangles $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$, each having area $\Delta A = \Delta x \Delta y$.

2. Choose a sample point (x_{ij}^*, y_{ij}^*) in each R_{ij} .

Take $f(x_{ij}^*, y_{ij}^*)$ as the height of a thin rectangular solid erected over R_{ij} ; its volume is $f(x_{ij}^*, y_{ij}^*) \Delta A$.

3. The sum of the volume of all these small rectangular solids approximates the volume of the solid under the graph of $z = f(x, y)$ over R . This sum

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

is called a **(double) Riemann sum** of f , and is used as an approximation of the value of the double integral.

We define the double integral of f over R as the limit of the Riemann sum by taking $m, n \rightarrow \infty$:

Definition 5.1. The **double integral** of f over a rectangle $R \subseteq \mathbb{R}^2$ is

$$\iint_R f(x, y) \, dA := \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

provided the limit exists and is the same for any choice of the sample points (x_{ij}^*, y_{ij}^*) in R_{ij} , for $1 \leq i \leq m$, $1 \leq j \leq n$. In this case, we say that f is **integrable**.

Theorem 5.2. All continuous functions are integrable.

If $f(x, y) \geq 0$, then the volume V of the solid that lies above a region D and below the surface $z = f(x, y)$ is the double integral

$$V = \iint_D f(x, y) \, dA.$$

The evaluation of double integrals from first principles is difficult, but here we see how to express a double integral as an iterated integral, which can then be evaluated by calculating two single integrals.

Definition 5.3. Let $f(x, y)$ be defined on $R = [a, b] \times [c, d]$. We write

$$\int_c^d f(x, y) dy$$

to mean that x is held fixed, and $f(x, y)$ is integrated with respect to y from c to d . Hence, $\int_c^d f(x, y) dy$ is a function of x , so we can integrate it with respect to x from a to b :

$$\int_a^b \int_c^d f(x, y) dy dx.$$

The resulting integral is called an **iterated integral**.

Similarly, one can define the iterated integral

$$\int_c^d \int_a^b f(x, y) dx dy$$

where we first integrate with respect to x from a to b (keeping y fixed), and then with respect to y from c to d .

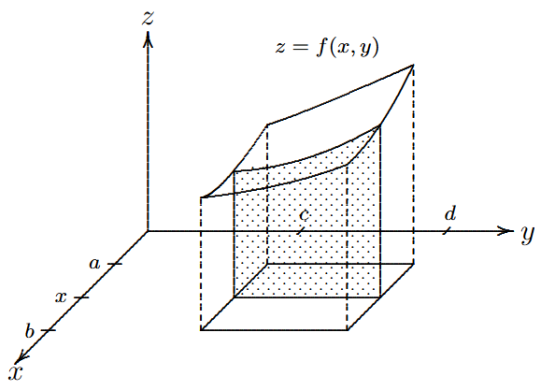


Figure 5.1: $\int_a^b \int_c^d f(x, y) dy dx$.

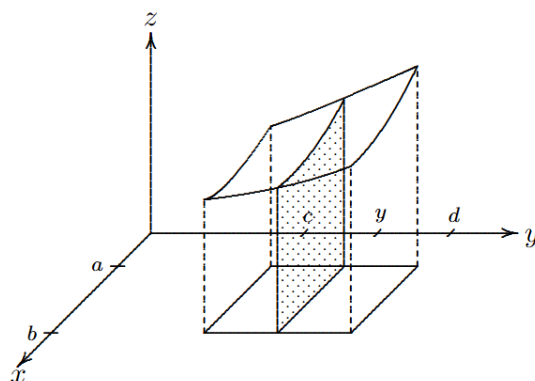


Figure 5.2: $\int_c^d \int_a^b f(x, y) dx dy$.

Theorem 5.4 (Fubini's theorem). If $f(x, y)$ is continuous on $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy. \quad (25)$$

Remark. More generally, Fubini's theorem holds if f is bounded on R , f is discontinuous only at a finite number of smooth curves, and the iterated integrals exist. Furthermore, the theorem is valid for a general closed and bounded region.

In particular, if $f(x, y) = g(x)h(y)$ can be factored as the product of a function of x only and a function of y only, then the double integral of f can be written as

$$\iint_R f(x, y) dA = \int_a^b g(x) dx \int_c^d h(y) dy.$$

5.1.2 Double integrals over general regions

We now consider regions $D \subseteq \mathbb{R}^2$ which have a "good shape":

Definition 5.5. A plane region $D \subseteq \mathbb{R}^2$ is of **type I** if it lies between the graphs of two continuous functions of x :

$$D = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

where $g_1(x)$ and $g_2(x)$ are continuous on $[a, b]$.

Theorem 5.6. If f is continuous on a domain D of Type I, then

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

Tips: For Type I region, it is helpful to draw a **vertical** arrow which starts at the lower boundary $y = g_1(x)$ and ends at the upper boundary $y = g_2(x)$. This corresponds to the inner integral.

Definition 5.7. A planar region $D \subseteq \mathbb{R}^2$ is of **type II** if it lies between the graphs of two continuous functions of y :

$$D = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

where $h_1(y)$ and $h_2(y)$ are continuous on $[c, d]$.

Theorem 5.8. If f is continuous on a domain D of Type II, then

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy.$$

Tips: For Type II region, it is helpful to draw a **horizontal** arrow which starts at the left boundary $x = h_1(y)$ and ends at the right boundary $x = h_2(y)$. This corresponds to the inner integral.

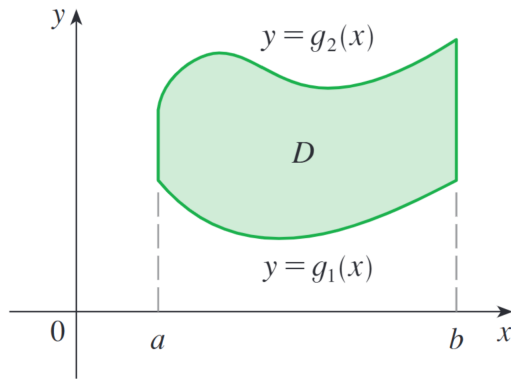


Figure 5.3: Type I region.

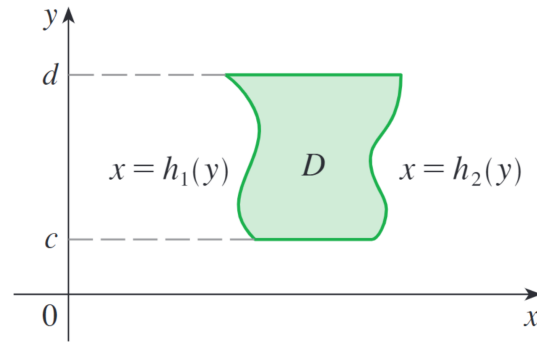


Figure 5.4: Type II region.

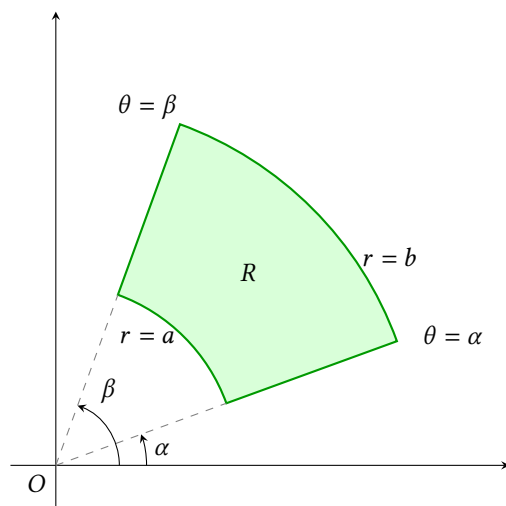
Lemma 5.9 (Properties of double integrals).

- (i) $\iint_D (f(x, y) + g(x, y)) \, dA = \iint_D f(x, y) \, dA + \iint_D g(x, y) \, dA.$
- (ii) $\iint_D c f(x, y) \, dA = c \iint_D f(x, y) \, dA$, where c is a constant.
- (iii) If $f(x, y) \geq g(x, y)$ for all $(x, y) \in D$, then $\iint_D f(x, y) \, dA \geq \iint_D g(x, y) \, dA.$
- (iv) $\iint_D f(x, y) \, dA = \iint_{D_1} f(x, y) \, dA + \iint_{D_2} f(x, y) \, dA$, where $D = D_1 \cup D_2$ and D_1, D_2 do not overlap except possibly at their boundary.
- (v) $\iint_D 1 \, dA = A(D)$, the area of D .
- (vi) If $m \leq f(x, y) \leq M$ for all $(x, y) \in D$, then $mA(D) \leq \iint_D f(x, y) \, dA \leq MA(D).$

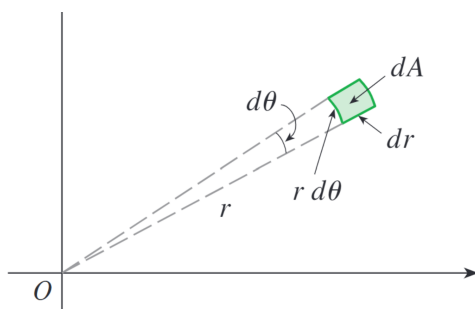
5.1.3 Double integrals in polar coordinates

Definition 5.10. A **polar rectangle** is a region

$$R = \{(r, \theta) : 0 \leq a \leq r \leq b, \alpha \leq \theta \leq \beta\}.$$



Consider a point (r, θ) on the plane in polar coordinates. An increment dr in r and $d\theta$ in θ give rise to an area $dA = r d\theta dr$. This is the area differential in polar coordinates.



Theorem 5.11 (Change to polar coordinates). Let f be a continuous function defined on a polar rectangle R given by

$$R = \{(r, \theta) : 0 \leq a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

where $0 \leq \beta - \alpha \leq 2\pi$. Then

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta. \quad (26)$$

When converting to polar coordinates:

1. Write $x = r \cos \theta$, $y = r \sin \theta$
2. Use the appropriate limits of integration for r and θ
3. Replace dA by $r dr d\theta$ (do not forget the additional r)

Next, we want to extend polar rectangles to more general **polar regions**. Similar to double integrals using rectangular coordinates, there are two types of polar regions.

Definition 5.12. A polar region of **Type I** is of the form

$$D = \{(r, \theta) : a \leq r \leq b, g_1(r) \leq \theta \leq g_2(r)\}$$

Theorem 5.13. If f is continuous on a polar region of type I

$$D = \{(r, \theta) : a \leq r \leq b, g_1(r) \leq \theta \leq g_2(r)\}$$

then

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(r)}^{g_2(r)} f(r \cos \theta, r \sin \theta) r \, d\theta \, dr. \quad (27)$$

Definition 5.14. A polar region of **type II** is of the form

$$D = \{(r, \theta) : \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

Theorem 5.15. If f is continuous on a polar region of type II

$$D = \{(r, \theta) : \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

then

$$\iint_D f(x, y) \, dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta. \quad (28)$$

5.2 Triple integrals

5.2.1 Triple integrals over boxes

Let $f: B \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function, where

$$B = [a, b] \times [c, d] \times [r, s]$$

is a rectangular box. We want to define the triple integral $\iiint_B f(x, y, z) \, dV$.

1. Divide B into sub-boxes by

- dividing $[a, b]$ into l subintervals $[x_{i-1}, x_i]$ of equal length Δx
- dividing $[c, d]$ into m subintervals $[y_{j-1}, y_j]$ of equal length Δy
- dividing $[r, s]$ into n subintervals $[z_{k-1}, z_k]$ of equal length Δz

Thus, B is divided into lmn sub-boxes B_{ijk} , where $1 \leq i \leq l, 1 \leq j \leq m, 1 \leq k \leq n$. Each sub-box has volume $\Delta V = \Delta x \Delta y \Delta z$.

2. Form the **(triple) Riemann sum**

$$\sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

where the sample point $(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \in B_{ijk}$.

Definition 5.16. The **triple integral** of f over B is defined as

$$\iiint_B f(x, y, z) \, dV := \lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

if the limit exists, and is independent of the choice of the partition and the sample points. In this case, we say that f is **integrable**.

Theorem 5.17. If f is continuous, then f is integrable.

Just as for double integrals, the practical method for evaluating triple integrals is to express them as iterated integrals:

Theorem 5.18 (Fubini's theorem). If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f(x, y, z) \, dV = \int_r^s \int_c^d \int_a^b f(x, y, z) \, dx \, dy \, dz. \quad (29)$$

Furthermore, the iterated integral may be evaluated in any order.

There are a total of $3! = 6$ possible orders in which we can integrate, all of which give the same value.

5.2.2 Triple integrals over general regions

Now we define the triple integral over a general bounded region E in three-dimensional space (a solid) by much the same procedure that we used for double integrals.

Definition 5.19.

- A solid region E is of **type I** if it lies between the graphs of two continuous functions of x and y .
- A solid region E is of **type II** if it lies between the graphs of two continuous functions of y and z .
- A solid region E is of **type III** if it lies between the graphs of two continuous functions of x and z .

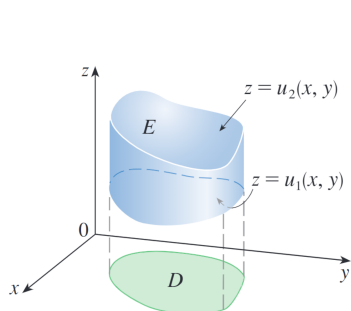


Figure 5.5: Type I region.

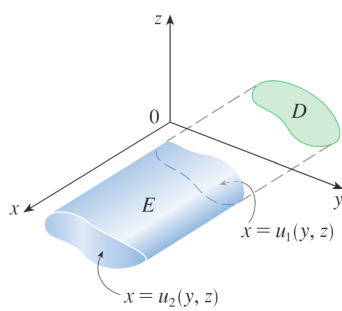


Figure 5.6: Type II region.

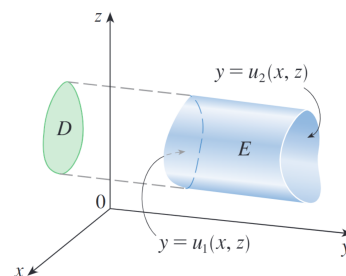


Figure 5.7: Type III region.

For each type of solid region, we can write the triple integral as an iterated integral of a double integral and a simple integral:

$$\begin{aligned}\iiint_E f(x, y, z) \, dV &= \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \right] dA \\ \iiint_E f(x, y, z) \, dV &= \iint_D \left[\int_{u_1(y, z)}^{u_2(y, z)} f(x, y, z) \, dx \right] dA \\ \iiint_E f(x, y, z) \, dV &= \iint_D \left[\int_{u_1(x, z)}^{u_2(x, z)} f(x, y, z) \, dy \right] dA\end{aligned}$$

For type 1 region, think of the inner integral with respect to z as a vertical strip. Then take vertical strips over $(x, y) \in D$ to get the entirety of E .

Triple integrals can be used to calculate volume of solids: the triple integral of $f(x, y, z) = 1$ over E gives the volume V of E :

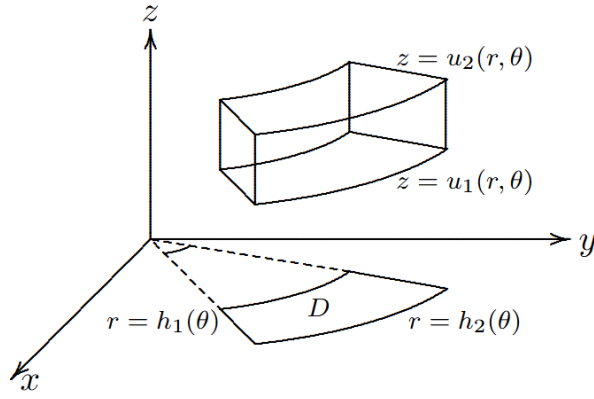
$$V = \iiint_E dV.$$

5.2.3 Triple integrals in cylindrical coordinates

When computing $\iiint_E f(x, y, z) \, dV$, we prefer cylindrical coordinates if the projection D of the solid E onto the xy -plane can be conveniently described in terms of polar coordinates.

Consider a rectangle in cylindrical coordinates:

$$E = \{(r, \theta, z) : \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta), u_1(r, \theta) \leq z \leq u_2(r, \theta)\}.$$



The volume element in cylindrical coordinates is $dV = r \, dz \, dr \, d\theta$.

Theorem 5.20. *The triple integral of $f(x, y, z)$ over E is*

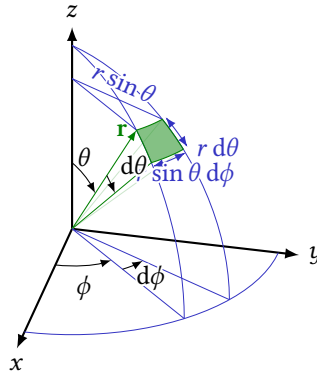
$$\begin{aligned} \iiint_E f(x, y, z) \, dV &= \iint_D \left[\int_{u_1(r, \theta)}^{u_2(r, \theta)} f(r \cos \theta, r \sin \theta, z) \, dz \right] dA \\ &= \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r, \theta)}^{u_2(r, \theta)} f(r \cos \theta, r \sin \theta, z) r \, dz \, dr \, d\theta. \end{aligned} \quad (30)$$

5.2.4 Triple integrals in spherical coordinates

Spherical coordinates simplify equations for regions involving spheres or cones. Hence, we use spherical coordinates to simplify triple integrals involving spheres or cones.

Consider the volume element in spherical coordinates. To do so, take any point $P(\rho, \theta, \phi)$, and make an increment in each of the coordinates.

We calculate the volume of the solid arising from these increments. The projection of OP onto the xy -plane has length $\rho \sin \phi$. Thus, the thickness of this volume element is $\rho \sin \phi \, d\theta$. It opens up a sector of width of $\rho \, d\phi$. Hence, the volume is $dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$.



Now consider a spherical rectangular solid

$$E = \{(\rho, \theta, \phi) : a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

where $a \geq 0$, $\beta - \alpha \leq 2\pi$, $d - c \leq \pi$.

Theorem 5.21. *The triple integral of f over E is*

$$\iiint_E f(x, y, z) \, dV = \int_c^d \int_{\alpha}^{\beta} \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi. \quad (31)$$

This formula can be extended to include more general spherical region such as

$$E = \{(\rho, \theta, \phi) : c \leq \rho \leq d, \alpha \leq \theta \leq \beta, g_1(\theta, \phi) \leq \phi \leq g_2(\theta, \phi)\}.$$

Theorem 5.22.

$$\iiint_E f(x, y, z) \, dV = \int_c^d \int_\alpha^\beta \int_{g_1(\theta, \phi)}^{g_2(\theta, \phi)} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi. \quad (32)$$

5.3 Change of variables

Changing from rectangular coordinates to polar, or cylindrical, or spherical coordinates, are special cases of a general process known as a **change of variables** or **transformation**.

A change of variables should be considered in any situation where we are presented with an integral that is difficult to evaluate in rectangular coordinates.

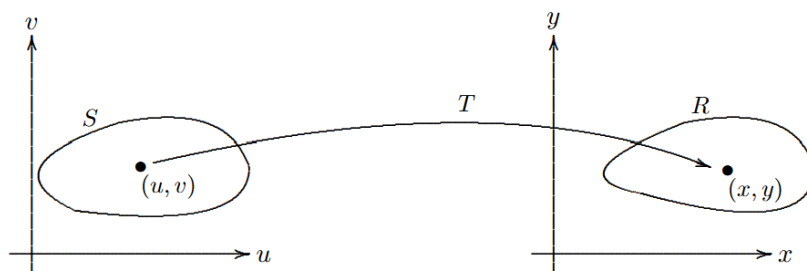
Definition 5.23. A **transformation** $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ from the uv -plane to the xy -plane is a map of the form

$$T(u, v) = (x(u, v), y(u, v)).$$

That is, x and y are functions of u and v :

$$x = x(u, v), \quad y = y(u, v).$$

In general, T transforms a region S in the uv -plane into a region R in the xy -plane, which is called the **image** of S , consisting of the images of all points in S .



Given a closed and bounded region S , how do we determine its image R under transformation T ?

The key is to examine the boundary. T maps the **boundary** of the domain into the **boundary** of the image. If S is closed and bounded, then R is closed and bounded. Hence, once we know the boundary of the image, we will be able to determine the image R .

Given $T: (u, v) \rightarrow (x, y)$, if S is a rectangle on the uv -plane, what is the area of the image of S under T ?

Definition 5.24. The **Jacobian** of the transformation T given by $x = x(u, v)$, $y = y(u, v)$ is

$$\frac{\partial(x, y)}{\partial(u, v)} := \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}.$$

Let dA be the image of the rectangle $du \, dv$ under the plane transformation T given by $x = x(u, v)$, $y = y(u, v)$. Then the Jacobian is the scaling factor:

$$dA = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv.$$

Theorem 5.25. Suppose $T(u, v)$ is a space transformation that maps a region S in the uv -plane onto a region R of

the xy -plane. Then

$$\iint_R f(x, y) \, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv . \quad (33)$$

Procedure:

1. Come up with a transformation: Define $u = u(x, y)$ and $v = v(x, y)$.
2. Solve for x and y in terms of u and v , i.e., $x = x(u, v)$ and $y = y(u, v)$.
3. Find the new region S , by substituting the boundary equations of R with your new variables to find the limits for u and v .
4. Calculate the Jacobian.
5. Substitute and evaluate the integral.

For the case of triple integrals, we have an analogous formula for change of variables.

Definition 5.26. Suppose $T(u, v, w) = (x(u, v, w), y(u, v, w), z(u, v, w))$ is a space transformation. The **Jacobian** of T is defined as

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} .$$

Intuitively, the transformation T scales

$$dV = dx \, dy \, dz = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du \, dv \, dw .$$

With this, we can compute a triple integral in x, y, z by changing it to a triple interval in u, v, w .

Theorem 5.27. If $f(x, y, z)$ is a continuous function defined on R , then

$$\iiint_R f(x, y, z) \, dV = \iiint_S f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du \, dv \, dw . \quad (34)$$

6 Vector Calculus

6.1 Vector fields

Definition 6.1. Let $D \subseteq \mathbb{R}^2$. A **vector field** on D is a function $\mathbf{F}$ that assigns to each point $(x, y) \in D$ a two-dimensional vector $\mathbf{F}(x, y)$.

We may write $\mathbf{F}(x, y)$ in terms of its component functions:

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j} = \langle P(x, y), Q(x, y) \rangle$$

or simply $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$.

Definition 6.2. Let $E \subseteq \mathbb{R}^3$. A **vector field** on E is a function $\mathbf{F}$ that assigns to each point $(x, y, z) \in E$ a three-dimensional vector $\mathbf{F}(x, y, z)$.

Similarly, in terms of component functions,

$$\begin{aligned}\mathbf{F}(x, y, z) &= P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k} \\ &= \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle\end{aligned}$$

or simply $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$.

Definition 6.3. A vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ defined on $D \subseteq \mathbb{R}^2$ is **continuous** if P and Q are continuous on D . Similarly, $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ defined on $E \subseteq \mathbb{R}^3$ is **continuous** if P , Q and R are continuous on E .

Definition 6.4. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function. Then ∇f is a vector field on $\mathbb{R}^2$, called the **gradient vector field** of f .

Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a differentiable function. Then ∇f is a vector field on $\mathbb{R}^3$, called the **gradient vector field** of f .

6.2 Line integrals

6.2.1 Line integrals of scalar fields

Consider a plane curve C defined by $\mathbf{r}(t) = \langle x(t), y(t) \rangle$, $a \leq t \leq b$. Assume C is a smooth curve, meaning that $\mathbf{r}'(t) \neq \mathbf{0}$, and $\mathbf{r}'(t)$ is continuous for all t . Let $f(x, y)$ be a continuous function defined in a domain containing C . We want to define the line integral

$$\int_C f(x, y) \, ds.$$

We follow the same procedure as before:

1. Subdivide the curve C from $\mathbf{r}(a)$ to $\mathbf{r}(b)$ into n small arcs of (equal) length Δs_i , $i = 1, \dots, n$.
[include diagram]
2. Pick an arbitrary point (x_i^*, y_i^*) inside the i -th small arc and form the Riemann sum $\sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i$.
3. The line integral of f along C is the limit of this Riemann sum:

$$\int_C f(x, y) \, ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i.$$

Definition 6.5. The **line integral** of f along C is defined to be

$$\int_C f(x, y) \, ds := \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i,$$

provided this limit exists and is the same for every choice of (x_i^*, y_i^*) .

We can pull back the integral to an integral in terms of t using the parametrization $\mathbf{r}$. Recall that the arc length differential is given by $ds = \|\mathbf{r}'(t)\| dt$. Thus,

$$\begin{aligned}\int_C f(x, y) ds &= \int_a^b f(x(t), y(t)) \|\mathbf{r}'(t)\| dt \\ &= \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.\end{aligned}\tag{35}$$

In particular, if $f(x, y) = 1$, then $\int_C ds$ is the arc length of C .

Geometrically, if $f(x, y) \geq 0$, then the line integral $\int_C f(x, y) ds$ represents the area of the “fence” or “curtain” shown below, whose base is C and height above the point (x, y) is $f(x, y)$.

[diagram]

Definition 6.6. A **piecewise smooth curve** C is a finite union of smooth curves $C_1, C_2, \dots, C_n$, where the initial point of C_{i+1} is the terminal point of C_i , $i = 0, \dots, n-1$. In this case, we write $C = C_1 + C_2 + \dots + C_n$.

Then the line integral f along C is defined to be

$$\int_C f(x, y) ds := \int_{C_1} f(x, y) ds + \dots + \int_{C_n} f(x, y) ds.$$

Next we define two more line integrals:

Definition 6.7. Given a smooth curve $C: \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, $a \leq t \leq b$. Define

$$\begin{aligned}\int_C f(x, y) dx &:= \int_a^b f(x(t), y(t)) x'(t) dt, \\ \int_C f(x, y) dy &:= \int_a^b f(x(t), y(t)) y'(t) dt,\end{aligned}$$

which are called the line integrals of f along C with respect to x and y .

Sometimes, we refer to the original line integral of f along C as “with respect to arc length”.

We make the following abbreviation:

$$\int_C P(x, y) dx + Q(x, y) dy = \int_C P(x, y) dx + \int_C Q(x, y) dy.$$

A parametrization $\mathbf{r}(t) = \langle x(t), y(t) \rangle$, $t \in [a, b]$ determines an orientation of C . In other words, C is an oriented curve. Note that if we reverse the orientation of C , we obtain a curve with the opposite orientation of C . We denote this oriented curve by $-C$.

Note that because the sign of $x'(t)$ and $y'(t)$ reverses in $-C$, we have

$$\int_{-C} f(x, y) dx = - \int_C f(x, y) dx \quad \text{and} \quad \int_{-C} f(x, y) dy = - \int_C f(x, y) dy.$$

But because the arc length differential is always positive,

$$\int_C f(x, y) ds = \int_{-C} f(x, y) ds.\tag{36}$$

For line integral of a function of three variables $f(x, y, z)$ along a parametrized space curve C , we have a similar

definition:

$$\begin{aligned}\int_C f(x, y, z) \, ds &= \int_a^b f(x(t), y(t)) \|\mathbf{r}'(t)\| \, dt \\ &= \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt.\end{aligned}$$

6.2.2 Line integrals of vector fields

Definition 6.8. Let $\mathbf{F}$ be a continuous vector field defined on a domain containing a smooth curve C given by $\mathbf{r}(t)$, $t \in [a, b]$. The **line integral** of $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} := \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt.$$

Note that $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}$ as $\mathbf{r}'(t)$ changes sign in $-C$.

We now rewrite $\int_C \mathbf{F} \cdot d\mathbf{r}$ in component form. Suppose $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$, and C is the curve given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $t \in [a, b]$. Then

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &:= \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt \\ &= \int_a^b \langle P(\mathbf{r}(t)), Q(\mathbf{r}(t)), R(\mathbf{r}(t)) \rangle \cdot \langle x'(t), y'(t), z'(t) \rangle \\ &= \int_a^b P(\mathbf{r}(t))x'(t) \, dt + \int_a^b Q(\mathbf{r}(t))y'(t) \, dt + \int_a^b R(\mathbf{r}(t))z'(t) \, dt \\ &= \int_C P \, dx + Q \, dy + R \, dz.\end{aligned}$$

Remark. Sometimes, it is helpful to think of $\mathbf{F} \cdot d\mathbf{r}$ as $\langle P, Q, R \rangle \cdot \langle dx, dy, dz \rangle = P \, dx + Q \, dy + R \, dz$.

Sometimes a curve C is not smooth, but it is a union of finitely many smooth curves $C_1, \dots, C_n$. In such cases, we compute the line integrals as sum of the line integrals over $C_1, \dots, C_n$:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \dots + \int_{C_n} \mathbf{F} \cdot d\mathbf{r}.$$

6.3 Conservative vector fields

Definition 6.9. A vector field $\mathbf{F}$ is **conservative** if it is the gradient of some scalar function f :

$$\mathbf{F} = \nabla f.$$

We call f a **potential function** for $\mathbf{F}$.

How can we determine whether a vector field is conservative? By definition, if we can find a potential function f such that $\mathbf{F} = \nabla f$, then $\mathbf{F}$ is conservative. Alternatively, as given by the following results, we can also do this without finding the potential function!

Definition 6.10. A **simple** curve is a curve which does not intersect itself.

Definition 6.11. A **simply-connected** region in the plane is a connected region such that every simple closed curve in D encloses only points that are in D .

Simply speaking, an open and simply connected region D is a region on the xy -plane without any holes, and any two points in it can be joined by a path entirely in D , and that it does not contain any of its boundary points.

Theorem 6.12 (Test for conservative field). Let $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ be defined on an open and simply connected region $D \subseteq \mathbb{R}^2$. Suppose P and Q have continuous partial derivatives in D . Then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad (37)$$

if and only if $\mathbf{F}$ is conservative on D .

We state the above theorem for three-dimensions:

Theorem 6.13 (Test for conservative field). Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ be defined on an open and simply connected region $D \subseteq \mathbb{R}^3$. Suppose P , Q and R have continuous partial derivatives in D . Then

$$\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial z}, \quad \frac{\partial R}{\partial x} = \frac{\partial P}{\partial z}, \quad \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} \quad (38)$$

if and only if $\mathbf{F}$ is conservative on D .

As we will see later, this is equivalent to $\text{curl } \mathbf{F} = \mathbf{0}$.

6.4 Fundamental theorem for line integrals

Recall the fundamental theorem for calculus:

$$\int_a^b F'(x) \, dx = F(b) - F(a).$$

It has the following generalization in terms of line integrals:

Theorem 6.14 (Fundamental theorem for line integrals). Let C be a smooth curve given by $\mathbf{r}(t)$, $t \in [a, b]$. Let f be a function of two or three variables whose gradient ∇f is continuous. Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a)). \quad (39)$$

This allows us to evaluate the line integral of a conservative vector field by only knowing the potential function at the endpoints.

Proof. Let $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where $t \in [a, b]$. Then, by chain rule,

$$\begin{aligned} \int_C \nabla f \cdot d\mathbf{r} &= \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt \\ &= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\ &= \int_a^b \frac{d}{dt} f(\mathbf{r}(t)) \, dt \\ &= f(\mathbf{r}(b)) - f(\mathbf{r}(a)). \end{aligned}$$

□

Corollary 6.15. The line integral of a conservative vector field is path independent.

Corollary 6.16. Suppose there are two paths with the same initial and terminal points, but their line integrals are different. Then the vector field is not conservative.

6.5 Green's theorem

Green's theorem relates a line integral along a simple closed curve C on the plane to the double integral over the region D bounded by C . This gives another method to evaluate line integrals.

The **positive orientation** of C refers a single counterclockwise traversal of C . Thus, the region D that it bounds is always on the left of C .

We use the notation ∂D to denote the boundary of D with the positive orientation.

Theorem 6.17 (Green's theorem). *Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane, and let D be the region bounded by C . If $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives on an open simply connected region that contains D , then*

$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA. \quad (40)$$

Though we use Green's theorem to evaluate a line integral by turning it into a double integral, sometimes we also use it to compute a double integral by turning it into a line integral.

One application is to compute areas of plane regions. To compute the area of D , given by $\iint_D dA$, using Green's theorem, we need to find P and Q such that

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1.$$

There are several candidates for P and Q :

- $P(x, y) = 0, Q(x, y) = x$.
- $P(x, y) = -y, Q(x, y) = 0$.
- $P(x, y) = -\frac{1}{2}y, Q(x, y) = \frac{1}{2}x$.

Theorem 6.18. *Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane. Let D be the region bounded by C . Then*

$$A(D) = \int_C x \, dy = - \int_C y \, dx = \frac{1}{2} \left(\int_C x \, dy - y \, dx \right). \quad (41)$$

6.6 Surface integrals

6.6.1 Parametric surfaces

Recall that a vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ of 1 degree of freedom, which traces out a curve in space. Now we consider a vector function of 2 degrees of freedom:

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D$$

which parametrizes a **surface** in $\mathbb{R}^3$ space. Then the following is a set of parametric equations for the surface S :

$$x = x(u, v), \quad y = y(u, v), \quad z = z(u, v).$$

6.6.2 Surface integrals of scalar fields

Given a scalar field $f(x, y, z)$ and a surface S , we want to define a surface integral

$$\iint_S f(x, y, z) \, dS.$$

Assume S is given by the vector equation

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D.$$

Assuming D is a rectangle, we divide it into subrectangles R_{ij} with dimensions $\Delta u \Delta v$. Then the surface S is divided into corresponding patches S_{ij} .

We evaluate f at a sample point P_{ij}^* in each patch, multiply the resulting value by the area ΔS_{ij} of the patch to form the (double) Riemann sum

$$\sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \Delta S_{ij}.$$

We define the surface integral by taking the limit of the above Riemann sum:

Definition 6.19. The surface integral of a scalar field is

$$\iint_S f(x, y, z) \, dS := \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \Delta S_{ij}$$

provided the limit exists and is the same for all choices of sample points P_{ij}^* .

For our purposes, we also require the parametric surface S to be **smooth**.

Definition 6.20. A surface S with parametrization $\mathbf{r}(u, v)$ is **smooth** if $\mathbf{r}_u$ and $\mathbf{r}_v$ are continuous, and $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$ for all $(u, v) \in D$.

Basically, a smooth surface is one which has no corners and no breaks.

With this, we arrive at the following formula for computing surface integral of scalar field by means of parametrization.

Theorem 6.21. Suppose S is parametrized by a vector function $\mathbf{r}(u, v)$ over D . Then the surface integral of $f(x, y, z)$ over S is

$$\iint_S f(x, y, z) \, dS = \iint_D f(x(u, v), y(u, v), z(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| \, dA. \quad (42)$$

If S is a union of finitely many smooth surfaces $S_1, \dots, S_n$ that intersect only along their boundaries, then the surface integral of f over S is the sum of the surface integrals over each S_i :

$$\iint_S f(x, y, z) \, dS = \iint_{S_1} f(x, y, z) \, dS + \dots + \iint_{S_n} f(x, y, z) \, dS.$$

In particular, if the surface S is given by $z = g(x, y)$, $(x, y) \in D$, then we can take x and y to be parameters:

$$\mathbf{r}(x, y) = \langle x, y, z \rangle = \langle x, y, g(x, y) \rangle.$$

Then $\mathbf{r}_x = \langle 1, 0, \frac{\partial g}{\partial x} \rangle$ and $\mathbf{r}_y = \langle 0, 1, \frac{\partial g}{\partial y} \rangle$, so that $\mathbf{r}_x \times \mathbf{r}_y = \langle -g_x, -g_y, 1 \rangle$. Hence, the surface integral is

$$\iint_S f(x, y, z) \, dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1} \, dA. \quad (43)$$

Theorem 6.22 (Surface area). Suppose S is a smooth parametric surface given by

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D$$

such that S is covered just once as (u, v) ranges throughout the domain D . Then the surface area of S is

$$A(S) = \iint_S dS = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| \, dA. \quad (44)$$

6.6.3 Surface integrals of vector fields

To define the surface integral over a vector field, we require the surface to be **orientable**, namely, it is two-sided.

Definition 6.23. A surface S is **orientable** if it is possible to define a unit normal vector $\mathbf{n}$ at each point (x, y, z) not on the boundary of the surface, such that $\mathbf{n}$ is a continuous function of (x, y, z) .

There are only two orientations of an orientable surface S , namely one for each side of the surface S which corresponds to the choice where all the unit normal vectors point away from that side of the surface.

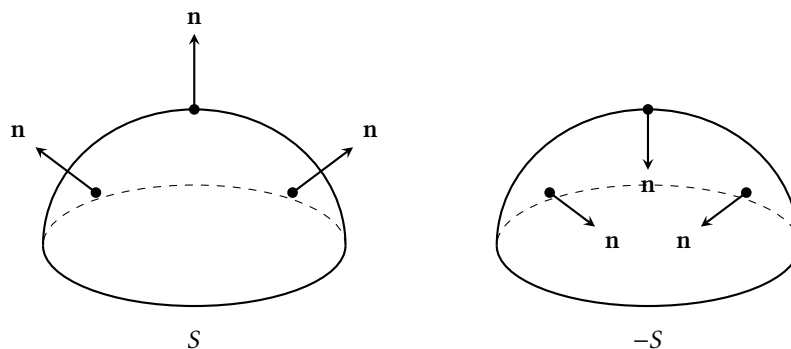


Figure 6.1: There are two possible orientations of S .

Such a unit normal vector can be found using the parametrization $\mathbf{r}(u, v)$ of the surface:

$$\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}.$$

The opposite orientation is denoted by $-\mathbf{n}$ and the corresponding oriented surface is denoted by $-S$.

In particular, if the surface S is given by $z = g(x, y)$, there is a natural orientation given by

$$\mathbf{n} = \frac{\mathbf{r}_x \times \mathbf{r}_y}{\|\mathbf{r}_x \times \mathbf{r}_y\|} = \frac{\langle -g_x, -g_y, 1 \rangle}{\sqrt{1 + g_x^2 + g_y^2}}.$$

As the $\mathbf{k}$ -component is positive, this gives the upward orientation of the surface.

Definition 6.24. Let $\mathbf{F}$ be a continuous vector field defined on an oriented surface S with unit normal vector $\mathbf{n}$. The **surface integral** of $\mathbf{F}$ over S is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} := \iint_S \mathbf{F} \cdot \mathbf{n} \, dS.$$

This integral is also called the **flux** of $\mathbf{F}$ over S .

Remark. Surface integrals of vector fields $\mathbf{F}$ are just surface integrals of a special kind of scalar function, namely $\mathbf{F} \cdot \mathbf{n}$ (which is related to the vector field $\mathbf{F}$ and oriented surface S given).

Remark. The integrand $\mathbf{F} \cdot \mathbf{n}$ (scalar function) gives (at any given point S) the component of $\mathbf{F}$ in the direction of $\mathbf{n}$.

In practice, we compute a surface integral as follows:

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, dS \\ &= \iint_S \mathbf{F} \cdot \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} \, dS \\ &= \iint_D \mathbf{F} \cdot \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} \|\mathbf{r}_u \times \mathbf{r}_v\| \, dA \\ &= \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA. \end{aligned}$$

Theorem 6.25 (Formula for Surface Integral of Vector Field).

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA. \quad (45)$$

In particular, if S is the graph of $z = g(x, y)$ over a region D in the xy -plane, and $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$, then the flux

across S in the **upward** orientation is

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, dS \\ &= \iint_D \langle P, Q, R \rangle \cdot \frac{\langle -g_x, -g_y, 1 \rangle}{\sqrt{g_x^2 + g_y^2 + 1}} \sqrt{g_x^2 + g_y^2 + 1} \, dA \\ &= \iint_D \left(-P \frac{\partial g}{\partial x} - Q \frac{\partial g}{\partial y} + R \right) dA.\end{aligned}$$

Similarly, the flux across S in the **downward** orientation is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left(P \frac{\partial g}{\partial x} + Q \frac{\partial g}{\partial y} - R \right) dA.$$

6.7 Divergence & curl

Definition 6.26. The **del operator** is

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

Definition 6.27. Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ be a vector field on $\mathbb{R}^3$. The **divergence** of $\mathbf{F}$ is

$$\operatorname{div} \mathbf{F} := \nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Definition 6.28. Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ be a vector field on $\mathbb{R}^3$. The **curl** of $\mathbf{F}$ is

$$\begin{aligned}\operatorname{curl} \mathbf{F} &:= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}.\end{aligned}$$

Remark. Unlike divergence, $\operatorname{curl} \mathbf{F}$ is a vector field.

Corollary 6.29. Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ be a vector field where P, Q, R have continuous second-order partial derivatives. Then

$$\operatorname{div}(\operatorname{curl} \mathbf{F}) = 0.$$

Proof. Compute

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{F}) &= \nabla \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \\ &= \nabla \cdot \left[\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} - \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \right] \\ &= \frac{\partial}{\partial x} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \\ &= \frac{\partial^2 R}{\partial x \partial y} - \frac{\partial^2 Q}{\partial x \partial z} + \frac{\partial^2 P}{\partial y \partial z} - \frac{\partial^2 R}{\partial y \partial x} + \frac{\partial^2 Q}{\partial z \partial x} - \frac{\partial^2 P}{\partial z \partial y} \\ &= \left(\frac{\partial^2 R}{\partial x \partial y} - \frac{\partial^2 R}{\partial y \partial x} \right) + \left(\frac{\partial^2 Q}{\partial z \partial x} - \frac{\partial^2 Q}{\partial x \partial z} \right) + \left(\frac{\partial^2 P}{\partial y \partial z} - \frac{\partial^2 P}{\partial z \partial y} \right) \\ &= 0 + 0 + 0 = 0,\end{aligned}$$

where the last step follows from Clairaut's theorem. $\square$

6.7.1 Divergence theorem

The divergence theorem relates the outward flux over a closed surface to a triple integral over the solid region bounded by that surface integral.

Theorem 6.30 (Divergence theorem). *Let E be a solid region whose boundary surface S is piecewise smooth with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region containing E . Then*

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \, dV. \quad (46)$$

This allows us to turn a surface integral into a triple integral.

6.7.2 Stokes' theorem

Stokes' theorem relates a line integral of vector field across a closed curve C to a surface integral over a surface whose boundary is C .

Let C be the boundary curve of an open surface S with unit normal $\mathbf{n}$. Imagine closing the fingers of your right hand such that your thumb is pointing generally in the direction of $\mathbf{n}$. Then the direction of your fingers represents the **positive orientation** of the curve C with respect to $\mathbf{n}$.

Theorem 6.31 (Stokes' theorem). *Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then*

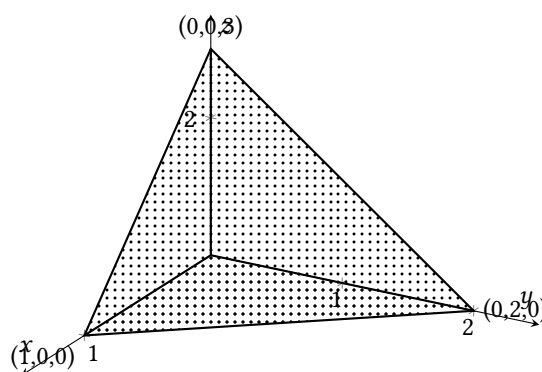
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot d\mathbf{S}. \quad (47)$$

A More Integrals from Tutorials

A.1 Tutorial 7

Problem A.1.1. Evaluate $\iiint_E xy \, dV$, where E is the solid tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 2, 0)$ and $(0, 0, 3)$.

Solution. Here E is the region bounded by the plane $6x + 3y + 2z = 6$, and the coordinate planes $x = 0$, $y = 0$ and $z = 0$. The x -, y -, and z -intercepts 1, 2 and 3 respectively.



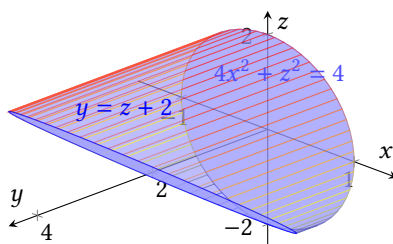
Each toothpick z ranges from 0 to $3 - 3x - 3y/2$. Therefore,

$$\begin{aligned} \iiint_E xy \, dV &= \int_0^1 \int_0^{2-2x} \int_0^{3-3x-3y/2} xy \, dz \, dy \, dx = \int_0^1 \int_0^{2-2x} (3xy - 3x^2y - \frac{3}{2}xy^2) \, dy \, dx \\ &= \int_0^1 \left[\frac{3}{2}xy^2 - \frac{3}{2}x^2y^2 - \frac{1}{2}xy^3 \right]_{y=0}^{y=2-2x} dx = \int_0^1 (2x - 6x^2 + 6x^3 - 2x^4) \, dx \\ &= \left[x^2 - 2x^3 + \frac{3}{2}x^4 - \frac{2}{5}x^5 \right]_0^1 = \frac{1}{10}. \end{aligned}$$

□

Problem A.1.2. Use a triple integral to find the volume of the solid bounded by the elliptic cylinder $4x^2 + z^2 = 4$ and the planes $y = 0$ and $y = z + 2$.

Solution.



Each toothpick y ranges from 0 to $z + 2$. Hence,

$$\begin{aligned} V &= \int_{-1}^1 \int_{-\sqrt{4-4x^2}}^{\sqrt{4-4x^2}} \int_0^{z+2} dy \, dz \, dx = 2 \int_0^1 \int_{-\sqrt{4-4x^2}}^{\sqrt{4-4x^2}} \int_0^{z+2} dy \, dz \, dx \\ &= 2 \int_0^1 \int_{-\sqrt{4-4x^2}}^{\sqrt{4-4x^2}} (z + 2) \, dz \, dx = 2 \int_0^1 \left[\frac{1}{2}z^2 + 2z \right]_{z=-2\sqrt{1-x^2}}^{z=2\sqrt{1-x^2}} dx \\ &= 2 \int_0^1 8\sqrt{1-x^2} \, dx = 16 \left[\frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x \right]_0^1 = 4\pi. \end{aligned}$$

□

Problem A.1.3. Evaluate the integral

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{2-x^2-y^2} (x^2+y^2)^{3/2} dz dy dx$$

by changing to cylindrical coordinates.

Solution. The region of integration E is the region above the paraboloid $z = x^2 + y^2$, or $z = r^2$, and below the paraboloid $z = 2 - x^2 - y^2$ or $z = 2 - r^2$.

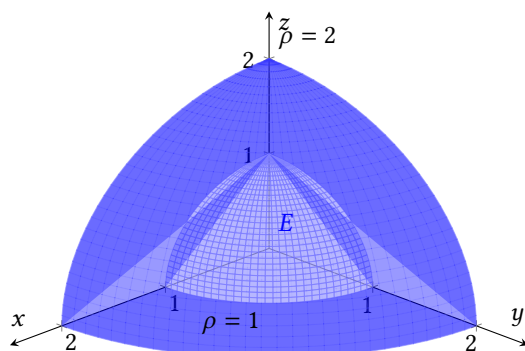
Also, the given bounds $-1 \leq x \leq 1$ and $-\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}$ precisely describe the unit circle in the xy -plane. Thus,

$$\begin{aligned} \int_0^{2\pi} \int_0^1 \int_{r^2}^{2-r^2} (r^2)^{3/2} r dz dr d\theta &= \int_0^{2\pi} \int_0^1 [r^4 z]_{z=r^2}^{z=2-r^2} dr d\theta \\ &= \int_0^{2\pi} \int_0^1 (2r^4 - 2r^6) dr d\theta \\ &= \int_0^{2\pi} \left(\frac{2}{5} - \frac{2}{7} \right) d\theta = \frac{8\pi}{35}. \end{aligned}$$

□

Problem A.1.4. Evaluate $\iiint_E z dV$, where E is the region lying between the spheres $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 4$ in the first octant.

Solution. In spherical coordinates, the bounds of E are $1 \leq \rho \leq 2$, $0 \leq \theta \leq \pi/2$, $0 \leq \phi \leq \pi/2$.



Hence,

$$\begin{aligned} \iiint_E z dV &= \int_0^{\pi/2} \int_0^{\pi/2} \int_1^2 (\rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi \\ &= \int_0^{\pi/2} \cos \phi \sin \phi d\phi \int_0^{\pi/2} d\theta \int_1^2 \rho^3 d\rho \\ &= \frac{1}{2} \cdot \frac{\pi}{2} \cdot \frac{15}{4} = \frac{15\pi}{16}. \end{aligned}$$

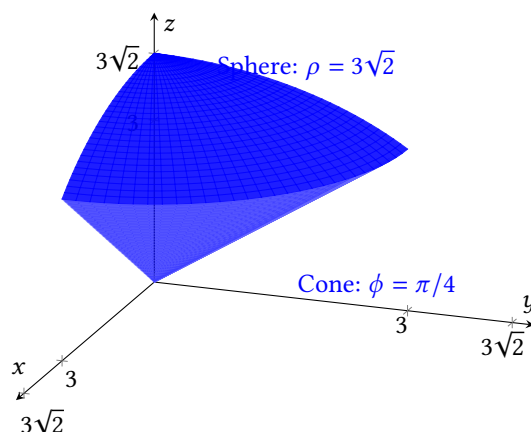
□

Problem A.1.5. Evaluate the integral

$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}} (x^2+y^2+z^2) dz dx dy$$

by changing to spherical coordinates.

Solution. The region of integration E is the region above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 18$ in the first octant.



Since E is in the first octant, we have $0 \leq \theta \leq \frac{\pi}{2}$. The cone has equation $\phi = \frac{\pi}{4}$ and so $0 \leq \phi \leq \frac{\pi}{4}$. Also $0 \leq \rho \leq \sqrt{18} = 3\sqrt{2}$.

Therefore, the integral becomes

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \int_0^{3\sqrt{2}} \rho^2 (\rho^2 \sin \phi) d\rho d\phi d\theta &= \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{4}} \sin \phi d\phi \int_0^{3\sqrt{2}} \rho^4 d\rho \\ &= \frac{\pi}{2} \cdot \left(1 - \frac{\sqrt{2}}{2}\right) \cdot \frac{972\sqrt{2}}{5} = \frac{486\pi(\sqrt{2}-1)}{5}. \end{aligned}$$

□

Problem A.1.6. Find the volume of the wedge in the first octant that is cut from the cylinder $y^2 + z^2 = 1$ by the planes $y = x$ and $x = 1$.

Solution. The wedge can be described as the region: $D = \{(x, y, z) | y^2 + z^2 \leq 1, 0 \leq x \leq 1, 0 \leq y \leq x\} = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq x, 0 \leq z \leq \sqrt{1-y^2}\}$. So the integral expressing the volume of the wedge is

$$\begin{aligned} \iiint_D dV &= \int_0^1 \int_0^x \int_0^{\sqrt{1-y^2}} dz dy dx \\ &= \int_0^1 \int_0^x \sqrt{1-y^2} dy dx \\ &= \int_0^1 \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right] dx \\ &= \left[-\frac{1}{6} (1-x^2)^{\frac{3}{2}} + \frac{1}{2} x \sin^{-1} x + \frac{1}{2} (1-x^2)^{\frac{1}{2}} \right]_0^1 \\ &= \frac{\pi}{4} - \frac{1}{3}. \end{aligned}$$

□

Problem A.1.7. Using cylindrical coordinates, evaluate $\iiint_E \sqrt{x^2 + y^2} dV$, where E is the region that lies inside the cylinder $x^2 + y^2 = 16$ and between the planes $z = -5$ and $z = 4$.

Solution. In cylindrical coordinates, E is given by $\{(r, \theta, z) | 0 \leq \theta \leq 2\pi, 0 \leq r \leq 4, -5 \leq z \leq 4\}$. So

$$\begin{aligned} \iiint_E \sqrt{x^2 + y^2} dV &= \int_0^{2\pi} \int_0^4 \int_{-5}^4 \sqrt{r^2} r dz dr d\theta = \int_0^{2\pi} d\theta \int_0^4 r^2 dr \int_{-5}^4 dz \\ &= (2\pi) \left(\frac{64}{3}\right) (9) = 384\pi. \end{aligned}$$

□

Problem A.1.8. Evaluate the integral

$$\int_0^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} z^2 \, dz \, dy \, dx$$

by changing into spherical coordinates.

Solution. The region of integration is the solid hemisphere E with nonnegative x -coordinates. In spherical coordinates, $E = \{(\rho, \phi, \theta) | 0 \leq \rho \leq 1, 0 \leq \phi \leq \pi, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}\}$.

$$\begin{aligned} \int_0^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} z^2 \, dz \, dy \, dx &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{\pi} \int_0^1 (\rho \cos \phi)^2 (\rho^2 \sin \phi) \, d\rho \, d\phi \, d\theta \\ &= \frac{1}{5} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{\pi} \cos^2 \phi \sin \phi \, d\phi \, d\theta \\ &= \frac{1}{5} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[-\frac{1}{3} \cos^3 \phi \right]_0^{\pi} d\theta = \frac{2}{15} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta = \frac{2\pi}{15}. \end{aligned}$$

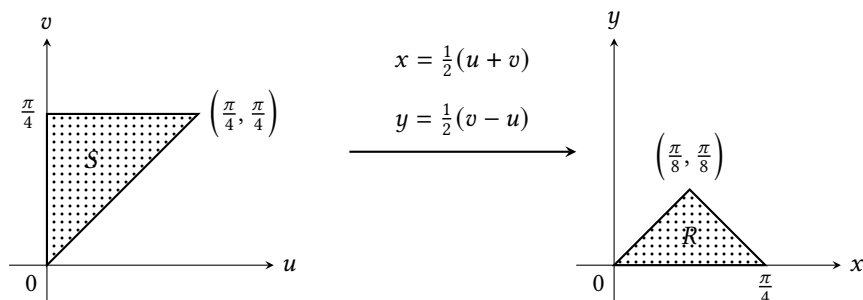
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Problem A.1.9. Use the change of variables $u = x - y$ and $v = x + y$, evaluate

$$\iint_R \frac{\sin(x-y)}{\cos(x+y)} \, dA,$$

where R is the triangular region bounded by the lines $y = 0$, $y = x$, and $x + y = \frac{\pi}{4}$.

Solution. Given the triangular region R has vertices $(0, 0)$, $(\frac{\pi}{4}, 0)$, $(\frac{\pi}{8}, \frac{\pi}{8})$. Under the change of variables $u = x - y$, $v = x + y$, it is transformed onto the triangular region S with vertices $(0, 0)$, $(\frac{\pi}{4}, \frac{\pi}{4})$, $(0, \frac{\pi}{4})$.



The transformation from S to R is

$$x = \frac{1}{2}(u+v), \quad y = \frac{1}{2}(-u+v)$$

where we solve for x and y in terms of u and v . The bounds of S are $0 \leq u \leq \frac{\pi}{4}$, $u \leq v \leq \frac{\pi}{4}$.

The Jacobian of this transformation is

$$\left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{vmatrix} = \frac{1}{4} - \left(-\frac{1}{4}\right) = \frac{1}{2}.$$

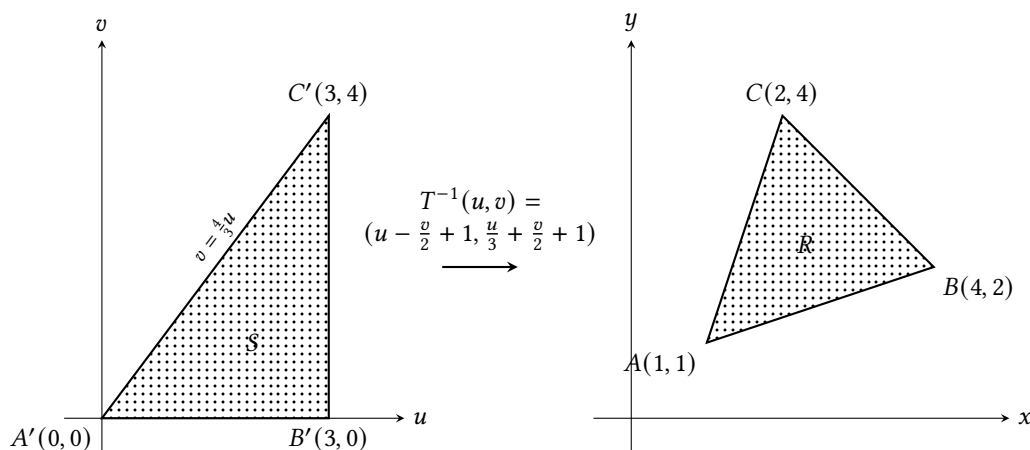
Therefore,

$$\begin{aligned}
 \iint_R \frac{\sin(x-y)}{\cos(x+y)} dA &= \iint_S \frac{\sin u}{\cos v} \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv = \int_0^{\pi/4} \int_0^v \frac{\sin u}{\cos v} \cdot \frac{1}{2} du dv \\
 &= \frac{1}{2} \int_0^{\pi/4} \left[-\frac{\cos u}{\cos v} \right]_{u=0}^{u=v} dv = \frac{1}{2} \int_0^{\pi/4} \left(-1 + \frac{1}{\cos v} \right) dv \\
 &= \frac{1}{2} \int_0^{\pi/4} (\sec v - 1) dv = \frac{1}{2} [\ln |\sec v + \tan v| - v]_{v=0}^{v=\pi/4} \\
 &= \frac{1}{2} \left[\ln(\sqrt{2} + 1) - \frac{\pi}{4} \right].
 \end{aligned}$$

□

Problem A.1.10. By using the transformation $T(x, y) = \left(\frac{3x}{4} + \frac{3y}{4} - \frac{3}{2}, -\frac{x}{2} + \frac{3y}{2} - 1 \right)$, evaluate the double integral $\iint_R (x - y) dA$, where R is the triangle in the xy -plane with vertices $A(1, 1)$, $B(4, 2)$ and $C(2, 4)$.

Solution. The transformation T maps bijectively the triangle R onto the triangle S with vertices $A'(0, 0)$, $B'(3, 0)$ and $C'(3, 4)$.



Let $u = 3x/4 + 3y/4 - 3/2$ and $v = -x/2 + 3y/2 - 1$. Then $x = u - v/2 + 1$, $y = u/3 + v/2 + 1$. Thus $T^{-1}(u, v) = (u - v/2 + 1, u/3 + v/2 + 1)$ maps S onto R . The Jacobian of T^{-1} is $\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 1 & -1/2 \\ 1/3 & 1/2 \end{vmatrix} = \frac{2}{3}$. Therefore,

$$\begin{aligned}
 \iint_R (x - y) dA &= \iint_S \frac{2}{3} (2u/3 - v) dA = \int_0^3 \int_0^{4u/3} \left(\frac{4u}{9} - \frac{2v}{3} \right) dv du \\
 &= \int_0^3 \left[\frac{4uv}{9} - \frac{v^2}{3} \right]_{v=0}^{v=4u/3} du = \int_0^3 0 du = 0.
 \end{aligned}$$

□

A.2 Tutorial 8

Problem A.2.1. Evaluate $\int_C x\sqrt{y} dx + 2y\sqrt{x} dy$, where C consists of the shortest arc of the circle $x^2 + y^2 = 1$ from $(1, 0)$ to $(0, 1)$ and the line segment joining $(0, 1)$ to $(4, 3)$.

Solution. Parametrize the arc C_1 of the circle $x^2 + y^2 = 1$ from $(1, 0)$ to $(0, 1)$ by $x = \cos t$, $y = \sin t$, $0 \leq t \leq \frac{\pi}{2}$.

Next, parametrize the line segment joining $(0, 1)$ to $(4, 3)$ by $x = 4t$, $y = 2t + 1$, $0 \leq t \leq 1$. Then

$$\begin{aligned} \int_C x\sqrt{y} \, dx + 2y\sqrt{x} \, dy &= \int_{C_1} x\sqrt{y} \, dx + 2y\sqrt{x} \, dy + \int_{C_2} x\sqrt{y} \, dx + 2y\sqrt{x} \, dy \\ &= \int_0^{\frac{\pi}{2}} [-\cos t (\sin t)^{\frac{1}{2}} \sin t + 2 \sin t (\cos t)^{\frac{1}{2}} \cos t] \, dt + \int_0^1 [16t\sqrt{2t+1} + 8(2t+1)] \, dt \\ &= \left[-\frac{2}{5}(\sin t)^{\frac{5}{2}} - \frac{4}{5}(\cos t)^{\frac{5}{2}} \right]_0^{\frac{\pi}{2}} + \left[\frac{16}{3}t(2t+1)^{\frac{3}{2}} - \frac{16}{15}(2t+1)^{\frac{5}{2}} + 8(t^2+t) \right]_0^1 \\ &= \frac{2}{5} + \frac{4}{5} + \frac{16}{3} \cdot 3\sqrt{3} - \frac{16}{15} \cdot 3^2\sqrt{3} + 16 - (0 - \frac{16}{15} + 0) \\ &= \frac{32\sqrt{3} + 66}{5}. \end{aligned}$$

[In the above, $\int_0^1 2t(2t+1)^{1/2} dt = \int_0^1 (2t+1)^{3/2} dt - \int_0^1 (2t+1)^{1/2} dt =$
 $= (1/2) \int_0^1 (2t+1)^{3/2} d(2t+1) - (1/2) \int_0^1 (2t+1)^{1/2} d(2t+1) = \dots]$ $\square$

Problem A.2.2. Evaluate $\int_C yz \, dx + xz \, dy + xy \, dz$, where C consists of the line segments from $(0, 0, 0)$ to $(2, 0, 0)$, from $(2, 0, 0)$ to $(1, 3, -1)$, and from $(1, 3, -1)$ to $(1, 3, 0)$.

Solution. Split $C = C_1 + C_2 + C_3$, and parametrize each line segment:

- Parametrize the line segment C_1 joining $(0, 0, 0)$ to $(2, 0, 0)$ by $x = 2t$, $y = z = 0$, where $0 \leq t \leq 1$.
- Parametrize the line segment C_2 joining $(2, 0, 0)$ to $(1, 3, -1)$ by $x = -t + 2$, $y = 3t$, $z = -t$, where $0 \leq t \leq 1$.
- Parametrize the line segment C_3 joining $(1, 3, -1)$ to $(1, 3, 0)$ by $x = 1$, $y = 3$, $z = t - 1$, where $0 \leq t \leq 1$.

Then

$$\int_C yz \, dx + xz \, dy + xy \, dz = 0 + \int_0^1 [3t^2 + 3(t^2 - 2t) - 3(2t - t^2)] \, dt + \int_0^1 3 \, dt = 0.$$

Alternatively, note that $\mathbf{F}$ is conservative, since $f(x, y, z) = xyz$ is a potential function. Hence, the above line integral is equal to $f(1, 3, 0) - f(0, 0, 0) = 0$. $\square$

Problem A.2.3. Let $\mathbf{F}(x, y, z) = 4xe^z\mathbf{i} + \cos y\mathbf{j} + 2x^2e^z\mathbf{k}$, and C the curve with vector equation given by $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^4\mathbf{k}$, $0 \leq t \leq 1$. Find a function f such that $\mathbf{F} = \nabla f$.

Hence, or otherwise, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Solution. Partially integrate $f_x(x, y, z) = 4xe^z$ with respect to x to get

$$f(x, y, z) = 2x^2e^z + g(y, z).$$

Then differentiate f with respect to y , and equate it to f_y :

$$g_y(y, z) = \cos y.$$

Integrate with respect to y to get

$$g(y, z) = \sin y + h(z).$$

Now $f(x, y, z) = 2x^2e^z + \sin y + h(z)$, so

$$f_z = 2x^2e^z + 0 + h'(z) = 2x^2e^z \implies h'(z) = 0 \implies h(z) = K$$

where K is a constant. We may take $K = 0$ to get $f(x, y, z) = 2x^2e^z + \sin y$ as the potential function.

The endpoints of the path $\mathbf{r}$ are $\mathbf{r}(0) = \langle 0, 0, 0 \rangle$ and $\mathbf{r}(1) = \langle 1, 1, 1 \rangle$. Therefore, by fundamental theorem of line integrals,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 1, 1) - f(0, 0, 0) = 2e + \sin 1.$$

□

Problem A.2.4. Let $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$, where $P(x, y) = \frac{-y}{x^2 + y^2}$ and $Q(x, y) = \frac{x}{x^2 + y^2}$.

- (a) Show that $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.
- (b) Show that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is not always independent of path.

Solution.

- (a) Using quotient rule, simply evaluate

$$\frac{\partial P}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad \frac{\partial Q}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2}.$$

- (b) Let C_1 be the upper semi-circular arc joining $(1, 0)$ to $(-1, 0)$, parametrized by $x = \cos t$, $y = \sin t$, for $0 \leq t \leq \pi$.

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^\pi \frac{(-\sin t)(-\sin t) + (\cos t)(\cos t)}{\cos^2 t + \sin^2 t} dt = \pi.$$

Let C_2 be the lower semi-circular arc joining $(1, 0)$ to $(-1, 0)$, parametrized by $x = \cos t$, $y = -\sin t$, for $0 \leq t \leq \pi$.

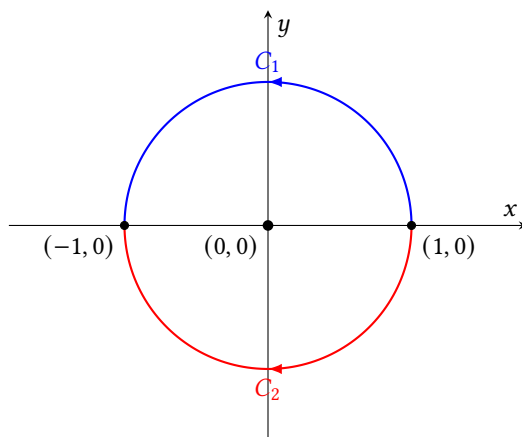
$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^\pi (-1) dt = -\pi.$$

Since these are not equal, the line integral of $\mathbf{F}$ is not independent of path.

Note that the domain of $\mathbf{F}$ is $\mathbb{R}^2 \setminus \{0\}$, which is not simply connected since it has a hole.

□

Remark. The two paths above were chosen to go around the hole, i.e., $(0, 0)$.

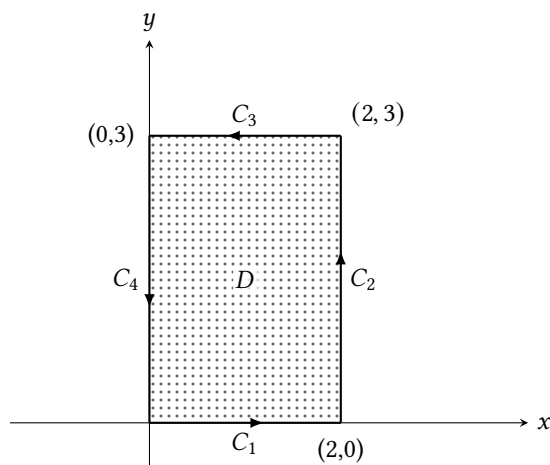


A.3 Tutorial 9

Problem A.3.1. Evaluate the line integral $\oint_C xy^2 dx + x^3 dy$, where C is the rectangle with vertices $(0, 0)$, $(2, 0)$, $(2, 3)$ and $(0, 3)$ by (a) direct computation, and (b) using Green's Theorem.

Solution.

- (a) Split $C = C_1 + C_2 + C_3 + C_4$:



Along C_1 : $x = t$, $y = 0$ for $0 \leq t \leq 2$.

Along C_2 : $x = 2$, $y = t$ for $0 \leq t \leq 3$.

Along C_3 : $x = 2 - t$, $y = 3$ for $0 \leq t \leq 2$.

Along C_4 : $x = 0$, $y = 3 - t$ for $0 \leq t \leq 3$.

Hence, the line integral is

$$\begin{aligned} \oint_C xy^2 dx + x^3 dy &= \int_{C_1+C_2+C_3+C_4} xy^2 dx + x^3 dy \\ &= \int_0^2 0 dt + \int_0^3 8 dt + \int_0^2 -9(2-t) dt + \int_0^3 0 dt \\ &= 0 + 24 - 18 + 0 = 6. \end{aligned}$$

(b) Use Green's theorem to evaluate the line integral:

$$\begin{aligned} \oint_C xy^2 dx + x^3 dy &= \iint_D \left[\frac{\partial}{\partial x}(x^3) - \frac{\partial}{\partial y}(xy^2) \right] dA = \int_0^2 \int_0^3 (3x^2 - 2xy) dy dx \\ &= \int_0^2 (9x^2 - 9x) dx = 6. \end{aligned}$$

□

Problem A.3.2. Evaluate

$$\oint_C (x^3 - y^3) dx + (x^3 + y^3) dy,$$

where C is the boundary of the region D between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$, where the orientation of C is so chosen that when you walk along C the region D is on your left.

Solution. By Green's Theorem applied to a multi-connected region, we have

$$\begin{aligned} \oint_C (x^3 - y^3) dx + (x^3 + y^3) dy &= \iint_{1 \leq x^2 + y^2 \leq 9} \left[\frac{\partial}{\partial x}(x^3 + y^3) - \frac{\partial}{\partial y}(x^3 - y^3) \right] dA \\ &= \iint_{1 \leq x^2 + y^2 \leq 9} (3x^2 + 3y^2) dA \\ &= 3 \int_0^{2\pi} \int_1^3 r^3 dr d\theta = 120\pi. \end{aligned}$$

□

Problem A.3.3. Find the area of the region bounded by the hypocycloid with vector equation

$$\mathbf{r}(t) = \cos^3 t \mathbf{i} + \sin^3 t \mathbf{j}, \quad 0 \leq t \leq 2\pi.$$

Solution. The area is given by

$$\begin{aligned} \oint_C x \, dy &= \int_0^{2\pi} (\cos^3 t)(3 \sin^2 t \cos t) \, dt = 3 \int_0^{2\pi} \cos^4 t \sin^2 t \, dt \\ &= 3 \int_0^{2\pi} \frac{1}{8} \left[\frac{1}{2} (1 - \cos 4t) \sin^2 2t \right] dt = \frac{3\pi}{8} \end{aligned}$$

where we used the double angle formula. □

Problem A.3.4. Let C be the closed curve oriented in the counter-clockwise sense with vector equation

$$\mathbf{r}(t) = 4 \cos t \mathbf{i} + 3 \sin^3 t \mathbf{j}, \quad 0 \leq t \leq 2\pi.$$

(i) Find the area of the region D enclosed by C .

(ii) Let $\mathbf{F}$ be the vector field on $\mathbb{R}^2$ defined by

$$\mathbf{F}(x, y) = (3xy^2 + 4 \cos x) \mathbf{i} + (3x^2y + 2x + e^y) \mathbf{j}.$$

Use Green's Theorem to evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Solution.

(i) The area of D is

$$\begin{aligned} \int_C x \, dy &= \int_0^{2\pi} (4 \cos t)(9 \sin^2 t)(\cos t) \, dt \\ &= \int_0^{2\pi} 36 \sin^2 t \cos^2 t \, dt = \int_0^{2\pi} 9 \sin^2(2t) \, dt \\ &= \frac{9}{2} \int_0^{2\pi} [1 - \cos(4t)] \, dt = \frac{9}{2} \left[t - \frac{1}{4} \sin(4t) \right]_0^{2\pi} = 9\pi. \end{aligned}$$

(ii) By Green's Theorem,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (3xy^2 + 4 \cos x) \, dx + (3x^2y + 2x + e^y) \, dy \\ &= \iint_D \left[\frac{\partial(3x^2y + 2x + e^y)}{\partial x} - \frac{\partial(3xy^2 + 4 \cos x)}{\partial y} \right] dA \\ &= \iint_D (6xy + 2 - 6xy) \, dA \\ &= \iint_D 2 \, dA = 2 \times \text{Area of } D = 18\pi. \end{aligned}$$

□

Problem A.3.5. Let $\mathbf{F}(x, y, z) = 2xy \mathbf{i} + (x^2 + 2yz) \mathbf{j} + y^2 \mathbf{k}$. Determine if $\mathbf{F}$ is a conservative vector field. If it is, find a function f such that $\nabla f = \mathbf{F}$.

Solution. To determine if $\mathbf{F}$ is conservative, we check if $\mathbf{F}$ is curl free.

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & x^2 + 2yz & y^2 \end{vmatrix} = (2y - 2y) \mathbf{i} - (0 - 0) \mathbf{j} + (2x - 2x) \mathbf{k} = \mathbf{0}.$$

Since $\mathbf{F}$ is defined on all of $\mathbb{R}^3$, and the partial derivatives of the component functions are continuous, we conclude that $\mathbf{F}$ is conservative.

To find the potential function f , first we know that $f_x(x, y, z) = 2xy$ so that $f(x, y, z) = x^2y + g(y, z)$ and $f_y(x, y, z) = x^2 + g_y(y, z)$. But $f_y(x, y, z) = x^2 + 2yz$, so $g_y(y, z) = 2yz$, which implies $g(y, z) = y^2z + h(z)$ and $f(x, y, z) = x^2y + y^2z + h(z)$. Thus, $f_z(x, y, z) = y^2 + h'(z)$. But $f_z(x, y, z) = y^2$ so $h'(z) = 0$, meaning $h(z) = K$ and $f(x, y, z) = x^2y + y^2z + K$, where K is a constant. $\square$

Problem A.3.6. Show that there does not exist a vector field $\mathbf{G}$ having continuous second order partial derivatives on $\mathbb{R}^3$ such that $\text{curl } \mathbf{G} = xy^2\mathbf{i} + yz^2\mathbf{j} + zx^2\mathbf{k}$.

Solution. Assume there exists such a vector field $\mathbf{G}$. Then $\text{div}(\text{curl } \mathbf{G}) = y^2 + z^2 + x^2 \neq 0$, a contradiction. $\square$

Problem A.3.7. Let $\mathbf{F}$ and $\mathbf{G}$ be vector fields in $\mathbb{R}^3$. Prove that

$$\text{div}(\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot \text{curl } \mathbf{F} - \mathbf{F} \cdot \text{curl } \mathbf{G}.$$

Solution. Let $\mathbf{F} = P_1\mathbf{i} + Q_1\mathbf{j} + R_1\mathbf{k}$ and $\mathbf{G} = P_2\mathbf{i} + Q_2\mathbf{j} + R_2\mathbf{k}$. Then

$$\begin{aligned} \text{div}(\mathbf{F} \times \mathbf{G}) &= \nabla \cdot (\mathbf{F} \times \mathbf{G}) = \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P_1 & Q_1 & R_1 \\ P_2 & Q_2 & R_2 \end{vmatrix} \\ &= \frac{\partial}{\partial x} \begin{vmatrix} Q_1 & R_1 \\ Q_2 & R_2 \end{vmatrix} - \frac{\partial}{\partial y} \begin{vmatrix} P_1 & R_1 \\ P_2 & R_2 \end{vmatrix} + \frac{\partial}{\partial z} \begin{vmatrix} P_1 & Q_1 \\ P_2 & Q_2 \end{vmatrix} \\ &= \left[Q_1 \frac{\partial R_2}{\partial x} + R_2 \frac{\partial Q_1}{\partial x} - Q_2 \frac{\partial R_1}{\partial x} - R_1 \frac{\partial Q_2}{\partial x} \right] - \left[P_1 \frac{\partial R_2}{\partial y} + R_2 \frac{\partial P_1}{\partial y} - P_2 \frac{\partial R_1}{\partial y} - R_1 \frac{\partial P_2}{\partial y} \right] \\ &\quad + \left[P_1 \frac{\partial Q_2}{\partial z} + Q_2 \frac{\partial P_1}{\partial z} - P_2 \frac{\partial Q_1}{\partial z} - Q_1 \frac{\partial P_2}{\partial z} \right] \\ &= \left[P_2 \left(\frac{\partial R_1}{\partial y} - \frac{\partial Q_1}{\partial z} \right) + Q_2 \left(\frac{\partial P_1}{\partial z} - \frac{\partial R_1}{\partial x} \right) + R_2 \left(\frac{\partial Q_1}{\partial x} - \frac{\partial P_1}{\partial y} \right) \right] \\ &\quad - \left[P_1 \left(\frac{\partial R_2}{\partial y} - \frac{\partial Q_2}{\partial z} \right) + Q_1 \left(\frac{\partial P_2}{\partial z} - \frac{\partial R_2}{\partial x} \right) + R_1 \left(\frac{\partial Q_2}{\partial x} - \frac{\partial P_2}{\partial y} \right) \right] \\ &= \mathbf{G} \cdot \text{curl } \mathbf{F} - \mathbf{F} \cdot \text{curl } \mathbf{G}. \end{aligned}$$

$\square$

A.4 Tutorial 10

Problem A.4.1. Find the area of the surface S consisting of the part of the sphere $x^2 + y^2 + z^2 = a^2$ that lies within the cylinder $x^2 + y^2 = ax$ and above the xy -plane.

Solution. Let $z = g(x, y) = \sqrt{a^2 - x^2 - y^2}$. Then

$$\begin{aligned} g_x &= -x(a^2 - x^2 - y^2)^{-1/2}, \\ g_y &= -y(a^2 - x^2 - y^2)^{-1/2}. \end{aligned}$$

Project S to xy -plane, you get the disc D bounded by the circle $x^2 + y^2 = ax$. Hence,

$$A(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2} \, dA.$$

The equation of the circle $(x - a/2)^2 + y^2 = (a/2)^2$ in polar coordinates is given by $r = a \cos \theta$. Therefore the

circular disk on the xy -plane bounded by the cylinder is given by

$$D = \{(r, \theta) : 0 \leq r \leq a \cos \theta, -\pi/2 \leq \theta \leq \pi/2\}.$$

Thus,

$$\begin{aligned} A(S) &= \iint_D \sqrt{\frac{x^2 + y^2}{a^2 - x^2 - y^2} + 1} \, dA = \int_{-\pi/2}^{\pi/2} \int_0^{a \cos \theta} \frac{a}{\sqrt{a^2 - r^2}} r \, dr \, d\theta \\ &= a \int_{-\pi/2}^{\pi/2} \left[-\sqrt{a^2 - r^2} \right]_0^{a \cos \theta} d\theta = a \int_{-\pi/2}^{\pi/2} (a - a|\sin \theta|) \, d\theta \\ &= a^2 [\theta + \cos \theta]_0^{\pi/2} - a^2 [\theta - \cos \theta]_{-\pi/2}^0 = a^2(\pi - 2). \end{aligned}$$

□

Problem A.4.2. Find a parametrisation of the portion S of the cone $z = \sqrt{x^2 + y^2}/3$ between the planes $z = 1$ and $z = 4/3$. Use the parametrisation to formulate the area of the surface as a double integral. Then, evaluate the integral.

Solution. In cylindrical coordinates, $(x, y, z) \mapsto (3z \cos \theta, 3z \sin \theta, z)$. Then S is parametrized by

$$\mathbf{r}(\theta, z) = \langle 3z \cos \theta, 3z \sin \theta, z \rangle,$$

where $0 \leq \theta \leq 2\pi$, $1 \leq z \leq \frac{4}{3}$. Then $\mathbf{r}_\theta = \langle -3z \sin \theta, 3z \cos \theta, 0 \rangle$ and $\mathbf{r}_z = \langle 3 \cos \theta, 3 \sin \theta, 1 \rangle$, so that

$$\mathbf{r}_\theta \times \mathbf{r}_z = \langle 3z \cos \theta, 3z \sin \theta, -9z \rangle.$$

The surface area is then

$$A(S) = \int_0^{2\pi} \int_1^{\frac{4}{3}} \|\mathbf{s}_\theta \times \mathbf{s}_z\| \, dz \, d\theta = \int_0^{2\pi} \int_1^{\frac{4}{3}} 3\sqrt{10}z \, dz \, d\theta = \frac{7\sqrt{10}}{3}\pi.$$

□

Problem A.4.3. Find a parametrization of the portion S of the plane $y + 2z = 2$ inside the cylinder $x^2 + y^2 = 1$. Use the parametrization to formulate the area of the surface S as a double integral. Then, evaluate the integral.

Solution. In cylindrical coordinates, $(x, y, z) \mapsto (r \cos \theta, r \sin \theta, z)$. Then we have $x^2 + y^2 \leq 1 \Rightarrow r \leq 1$ and $z = 1 - \frac{1}{2}y = 1 - \frac{r}{2} \sin \theta$. Hence, the surface can be parametrized as

$$\mathbf{s}(\theta, r) = \left\langle r \cos \theta, r \sin \theta, 1 - \frac{r}{2} \sin \theta \right\rangle$$

where $0 \leq r \leq 1$ and $0 \leq \theta \leq 2\pi$. Then the area is

$$\begin{aligned} A &= \iint dS = \int_0^1 \int_0^{2\pi} \|\mathbf{s}_\theta \times \mathbf{s}_r\| \, d\theta \, dr \\ &= \int_0^1 \int_0^{2\pi} \frac{\sqrt{5}}{2} r \, d\theta \, dr \\ &= \frac{\sqrt{5}}{2} \pi. \end{aligned}$$

Alternatively, the surface S projects to the unit disc $D : x^2 + y^2 \leq 1$ in the xy -plane, and S is parametrized by $z = g(x, y) = 1 - y/2$. Now $\langle -g_x, -g_y, 1 \rangle = \langle 0, 1/2, 1 \rangle$. Thus,

$$A = \iint_S 1 \, dS = \iint_D \sqrt{g_x^2 + g_y^2 + 1} \, dA = \frac{\sqrt{5}}{2} \times \text{Area}(D) = \frac{\sqrt{5}\pi}{2}.$$

□

Problem A.4.4. A torus of revolution (doughnut) is obtained by rotating a circle C in the xz plane about the z -axis. Suppose that C has a radius r and center $(R, 0, 0)$.

- (i) Find a parametrization $\mathbf{r}(u, v)$ of the torus. Specify the set D in which (u, v) must lie.

Hint: Let u be the angle that the line from the point $\mathbf{r}(u, v)$ on the torus to the center of the rotated circle form with the xy -plane, and let v be the angle formed by the line from the point $\mathbf{r}(u, v)$ on the torus to the origin with the positive x -axis.

- (ii) Show that the surface area of the torus is $4\pi^2 Rr$.

Solution.

- (i) First, consider the circle C in the xz -plane: $y = 0$ and $(x - R)^2 + z^2 = r^2$, which can be parametrized by $x - R = r \cos u$ and $z = r \sin u$, where $0 \leq u \leq 2\pi$. Then, the parametric equations for (x, z) on C is:

$$x = R + r \cos u, \quad z = r \sin u, \quad 0 \leq u \leq 2\pi.$$

We revolve the circle about the z -axis. Hence,

$$x = (R + r \cos u) \cos v, \quad y = (R + r \cos u) \sin v, \quad z = r \sin u,$$

where $0 \leq u \leq 2\pi$, $0 \leq v \leq 2\pi$. We can also write this as a vector function

$$\mathbf{r}(u, v) = (R + r \cos u) \cos v \mathbf{i} + (R + r \cos u) \sin v \mathbf{j} + r \sin u \mathbf{k},$$

where $(u, v) \in D = [0, 2\pi] \times [0, 2\pi]$.

- (ii) We first find $\|\mathbf{r}_u \times \mathbf{r}_v\|$:

$$\begin{aligned} \mathbf{r}_u &= \langle -r \sin u \cos v, -r \sin u \sin v, r \cos u \rangle, \\ \mathbf{r}_v &= \langle -(R + r \cos u) \sin v, (R + r \cos u) \cos v, 0 \rangle, \\ \mathbf{r}_u \times \mathbf{r}_v &= \langle -r(R + r \cos u) \cos u \cos v, -r(R + r \cos u) \cos u \sin v, -r(R + r \cos u) \sin u \rangle. \end{aligned}$$

Therefore, $\|\mathbf{r}_u \times \mathbf{r}_v\| = r(R + r \cos u)$. Then, the surface area of the torus is

$$\begin{aligned} \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| \, du \, dv &= \int_0^{2\pi} \int_0^{2\pi} r(R + r \cos u) \, du \, dv \\ &= \int_0^{2\pi} 2\pi Rr + r^2 \sin(2\pi) - r^2 \sin(0) \, dv \\ &= 2\pi Rr(2\pi) = 4\pi^2 Rr. \end{aligned}$$

□

Problem A.4.5. Evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for the vector field $\mathbf{F}(x, y, z) = z\mathbf{i} + y\mathbf{j} + x\mathbf{k}$, where S is the helicoid $\mathbf{r}(u, v) = \langle u \cos(v), u \sin(v), v \rangle$, $0 \leq u \leq 1$, $0 \leq v \leq \pi$, with upward orientation.

Solution. We have $\mathbf{r}_u \times \mathbf{r}_v = \langle \cos v, \sin v, 0 \rangle \times \langle -u \sin v, u \cos v, 1 \rangle = \langle \sin v, -\cos v, u \rangle$, and

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \int_0^\pi \int_0^1 \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, du \, dv \\ &= \int_0^\pi \int_0^1 \langle v, u \sin v, u \cos v \rangle \cdot \langle \sin v, -\cos v, u \rangle \, du \, dv \\ &= \int_0^\pi \int_0^1 (v \sin v - u \sin v \cos v + u^2 \cos v) \, du \, dv. \end{aligned}$$

Since both $\cos v$ and $\cos v \sin v$ have integral 0 from 0 to π , the last line becomes

$$\int_0^\pi \int_0^1 v \sin v \, du \, dv = \int_0^\pi v \sin v \, dv = - \int_0^\pi v (d \cos v) = \dots = \pi.$$

□

Problem A.4.6. Let $\mathbf{F}(x, y, z) = -y\mathbf{i} + x\mathbf{j} + 3z\mathbf{k}$. Evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where S is the hemisphere $z = \sqrt{16 - x^2 - y^2}$ with the upward orientation.

Solution. A parametrization of the hemisphere using spherical coordinates is

$$\mathbf{r}(\phi, \theta) = 4 \sin \phi \cos \theta \mathbf{i} + 4 \sin \phi \sin \theta \mathbf{j} + 4 \cos \phi \mathbf{k},$$

where $0 \leq \phi \leq \frac{\pi}{2}$, and $0 \leq \theta \leq 2\pi$.

Then, $\mathbf{r}_\phi \times \mathbf{r}_\theta = 16 \sin^2 \phi \cos \theta \mathbf{i} + 16 \sin^2 \phi \sin \theta \mathbf{j} + 16 \sin \phi \cos \phi \mathbf{k}$.

Also, $\mathbf{F}(\mathbf{r}(\phi, \theta)) = -4 \sin \phi \sin \theta \mathbf{i} + 4 \sin \phi \cos \theta \mathbf{j} + 12 \cos \phi \mathbf{k}$.

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F}(\mathbf{r}(\phi, \theta)) \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) \, d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/2} 192 \sin \phi \cos^2 \phi \, d\phi \, d\theta \\ &= 2\pi \left[-64 \cos^3 \phi \right]_0^{\pi/2} = 128\pi. \end{aligned}$$

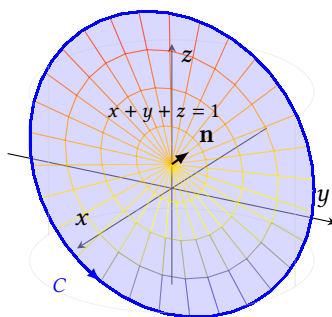
□

Problem A.4.7. Use Stokes' theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F}(x, y, z) = x^2 z \mathbf{i} + x y^2 \mathbf{j} + z^2 \mathbf{k}$$

and C is the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 9$ oriented counterclockwise as viewed from above.

Solution.



The curve of intersection is an ellipse C in the plane $x + y + z = 1$. Thus, C bounds an elliptical region D on this plane, and D is oriented by the unit normal vector $\mathbf{n} = \frac{1}{\sqrt{3}} \langle 1, 1, 1 \rangle$.

Now $\text{curl } \mathbf{F} = x^2 \mathbf{j} + y^2 \mathbf{k}$. For the plane $z = g(x, y) = 1 - x - y$, we have $g_x = -1$, $g_y = -1$ and $\langle -g_x, -g_y, 1 \rangle = \langle 1, 1, 1 \rangle$. Therefore, by Stokes' theorem,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_{x^2 + y^2 \leq 9} \langle x^2, y^2, 0 \rangle \cdot \langle 1, 1, 1 \rangle \, dx \, dy \\ &= \iint_{x^2 + y^2 \leq 9} (x^2 + y^2) \, dx \, dy = \int_0^{2\pi} \int_0^3 r^3 \, dr \, d\theta = \frac{81\pi}{2}. \end{aligned}$$

□

Problem A.4.8. Let $\mathbf{n}$ be the outer unit normal of the surface S given by $4x^2 + 9y^2 + 36z^2 = 26$, $z \geq 0$, and let $\mathbf{F}(x, y, z) = y\mathbf{i} + x^2\mathbf{j} + (x^2 + y^4)^{3/2} \sin(e^{\sqrt{xyz}})\mathbf{k}$. Find the value of

$$\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}.$$

Hint: One parametrization of the ellipse at the base of the shell is of the form $x = a \cos t$, $y = b \sin t$ for some constants a, b .

Solution. The surface S is bounded by the curve C which is the intersection of $4x^2 + 9y^2 + 36z^2 = 26$ and $z = 0$ (the xy -plane). Now $C : 4x^2 + 9y^2 = 26$ (positively oriented) is parametrized by

$$\left\langle \frac{\sqrt{26}}{2} \cos \theta, \frac{\sqrt{26}}{3} \sin \theta, 0 \right\rangle$$

where $0 \leq \theta \leq 2\pi$. Thus, we have

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \int_C \mathbf{F} \cdot d\mathbf{r} \\ &= \int_0^{2\pi} \left\langle \frac{\sqrt{26}}{3} \sin \theta, \frac{26}{4} \cos^2 \theta, \left(\frac{26}{4} \cos^2 \theta + \left(\frac{26}{9} \right)^2 \sin^4 \theta \right)^{3/2} \sin 1 \right\rangle \cdot \left\langle -\frac{\sqrt{26}}{2} \sin \theta, \frac{\sqrt{26}}{3} \cos \theta, 0 \right\rangle d\theta \\ &= \int_0^{2\pi} \left(-\frac{26}{6} \sin^2 \theta + \frac{26^{3/2}}{12} \cos^3 \theta \right) d\theta \\ &= -\frac{13}{3} \pi. \end{aligned}$$

□

Problem A.4.9. Let C be a simple closed smooth curve in the plane $x + 2y + z = 2$, oriented counterclockwise. Show that

$$\int_C 2y \, dx + 3z \, dy - x \, dz$$

depends only on the area of the region enclosed by C and not on the position or shape of C .

Solution. We use Stokes' theorem to evaluate the line integral. Let $\mathbf{F} = \langle 2y, 3z, -x \rangle$. Then $\text{curl } \mathbf{F} = \langle -3, 1, -2 \rangle$. Since C is on the plane $2x + 2y + z = 2$, the surface enclosed by C is on the plane and everywhere has unit normal vector $\frac{1}{3} \langle 2, 2, 1 \rangle$, so the surface area element is $d\mathbf{S} = \frac{\langle 2, 2, 1 \rangle}{3} dS$. Hence, by Stokes,

$$\int_C 2y \, dx + 3z \, dy - x \, dz = \int_C \mathbf{F} \cdot \langle dx, dy, dz \rangle = \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \langle -3, 1, -2 \rangle \cdot \frac{\langle 2, 2, 1 \rangle}{3} dS = -2 \times \text{Area of } S.$$

□

A.5 Tutorial 11

Problem A.5.1. Use the Divergence Theorem to calculate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where

$$\mathbf{F}(x, y, z) = x^3 \mathbf{i} + y^3 \mathbf{j} + z^3 \mathbf{k}$$

and S is the surface of the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = 2$.

Solution. Note that $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = 3(x^2 + y^2 + z^2)$. By the Divergence Theorem and using the cylindrical

coordinates (for the second equality below), we have

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} \, dV = \int_0^{2\pi} \int_0^1 \int_0^2 (3r^2 + 3z^2)r \, dz \, dr \, d\theta \\ &= 2\pi \int_0^1 (6r^3 + 8r) \, dr = 11\pi.\end{aligned}$$

□

Problem A.5.2. Use Gauss' divergence theorem to find the outward flux of $\mathbf{F}$ across the boundary of the region D , where $\mathbf{F}(x, y, z) = y\mathbf{i} + xy\mathbf{j} - z\mathbf{k}$ and D is the region inside the solid cylinder $x^2 + y^2 \leq 4$, between the plane $z = 0$ and the paraboloid $z = x^2 + y^2$.

Solution. We have $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = x - 1$. By Gauss' divergence theorem (and the cylindrical coordinates for the third equation below), we have

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_D \operatorname{div} \mathbf{F} \, dV \\ &= \int_0^{2\pi} \int_0^2 \int_0^{r^2} (r \cos \theta - 1)r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 (r^2 \cos \theta - r)z \Big|_0^{r^2} \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 (r^2 \cos \theta - r)r^2 \, dr \, d\theta \\ &= -8\pi.\end{aligned}$$

□

Problem A.5.3. Let

$$S_1 = \{(x, y, z) \mid z = \sqrt{x^2 + y^2}, 0 \leq z \leq 1/2\}$$

be the cone with the downward orientation and

$$S_2 = \{(x, y, z) \mid x^2 + y^2 + (z - \frac{1}{2})^2 = \frac{1}{4}, \frac{1}{2} \leq z \leq 1\}$$

be the hemisphere with the upward orientation. Let $S = S_1 + S_2$ be the closed surface with the outward orientation. Let $\mathbf{F}$ be the vector field on $\mathbb{R}^3$ given by

$$\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

(i) Evaluate $\iint_{S_1} \mathbf{F} \cdot d\mathbf{S}$.

(ii) Using the Divergence Theorem, or otherwise, show that

$$\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \frac{3\pi}{8}.$$

Solution.

(i) The cone and the hemisphere meet in a circle of radius $1/2$ at $z = 1/2$. Our S_1 has position vector $\mathbf{r}(x, y) = \langle x, y, \sqrt{x^2 + y^2} \rangle$. A downward normal vector of the cone is

$$\mathbf{n}_1 = -(\mathbf{r}_x \times \mathbf{r}_y) = \left\langle \frac{\partial}{\partial x} \sqrt{x^2 + y^2}, \frac{\partial}{\partial y} \sqrt{x^2 + y^2}, -1 \right\rangle = \left\langle \frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right\rangle.$$

On the cone S_1 , $\mathbf{F}(x, y, z) = \langle x, y, \sqrt{x^2 + y^2} \rangle$. Therefore,

$$\begin{aligned}\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= \iint_{x^2+y^2 \leq \frac{1}{4}} \mathbf{F} \cdot \mathbf{n}_1 dA \\ &= \iint_{x^2+y^2 \leq \frac{1}{4}} \left(x \cdot \frac{x}{\sqrt{x^2 + y^2}} + y \cdot \frac{y}{\sqrt{x^2 + y^2}} - \sqrt{x^2 + y^2} \right) dA = 0.\end{aligned}$$

(ii) Let E be the region bounded by S . By the Divergence Theorem,

$$\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} dV = 3 \iiint_E dV,$$

which is 3 times the volume of the solid region E . By (i), the first surface integral is 0. Thus,

$$\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = 3 \times \text{Volume of } E = 3 \left(\frac{1}{2} \cdot \frac{4\pi}{3} \left(\frac{1}{2} \right)^3 + \frac{1}{3} \pi \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)^2 \right) = \frac{3\pi}{8}.$$

□